Antwerp Management School
Master Thesis

From GenAI Practice Vacuum to Validated Guidance

A Delphi Study of AI-Augmented Reference Architecture Development

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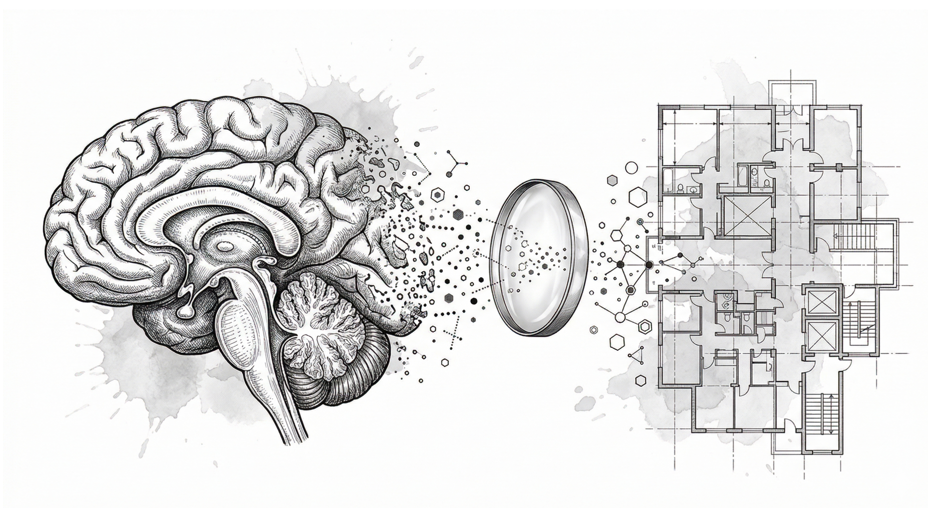

A thesis submitted in fulfilment of the requirements for the degree of Master of Enterprise IT Architecture (MSc)

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Executive Summary

Generative AI (GenAI) is changing how professionals work (Brynjolfsson et al., 2025; Noy & Zhang, 2023). Enterprise Architecture (EA) practitioners have started using GenAI to develop Reference Architectures (RAs), but no empirical research documents how they do so (Banh et al., 2025). Practitioners experiment in isolation. They lack shared vocabulary, evaluation criteria, or validated guidance. This study asks: *What practices do EA practitioners adopt when using GenAI for RA development, and how can these be systematized into guidance?*

We conducted a three-round Design-Oriented Delphi study (Skulmoski et al., 2007) with seven to ten EA practitioners per round. Round 1 (eight participants) collected asynchronous discovery workbooks. Round 2 (ten participants) convened synchronous sessions to validate emerging patterns. Round 3 (seven participants) refined verification heuristics and critiqued a draft guidance artifact.

The panel identified seven practice categories that span RA development activities. System/Codebase Analysis drew the strongest consensus. Stakeholder Communication proved most context-dependent. Three shifts describe how GenAI changes EA work. First, practitioners now spend more time checking GenAI outputs than producing initial content. Second, the practitioner role moves from creator to curator. Third, GenAI lets practitioners engage with topics beyond their immediate expertise, which changes how roles interact.

Practitioners decide how much to trust GenAI based on the task at hand. The dominant mental model treats GenAI as a junior colleague: capable, but requiring supervision. Eight verification heuristics, organized in two tiers (structural and domain), give practitioners concrete checks for validating outputs. Table 1 shows the resulting trust spectrum from full delegation to full human retention.

Two tensions cut across these findings. The first is a Ghost Architect dynamic: when GenAI enables anyone to produce architectural artifacts, it becomes unclear who is accountable. Organizations respond by shifting from periodic gate reviews to continuous guardrails. The second is a competency decay risk: automating foundational tasks (e.g., cold-start writing, exhaustive research) may erode the learning process through which architectural expertise develops. Practitioners counter this with deliberate practices such as AI-free design sessions and independent-attempt-first protocols.

This study extends distributed cognition theory (Hutchins, 1995). GenAI acts as a new kind of cognitive partner whose role is negotiated through practice, not predetermined by system design. On the practical side, the study provides a taxonomy of GenAI practices grounded in observable behavior. It shows where GenAI adds consistent value and where

Table 1. Trust Spectrum for GenAI in RA Development

Trust Level	**Task Type**	**Verification Approach**
Highest	Information compression: summaries, boilerplate, dependency graphs	Spot-check representative samples
High	Documentation drafts, concept explanation, test generation	Review structure and key claims
Medium	Code refactoring, common patterns, meeting synthesis	Verify against organizational standards
Medium-Low	Design trade-off exploration, technology recommendations	Cross-validate with primary sources
Low	Architecture decisions, novel business logic	Independent domain expert review
Very Low	Security-critical code, compliance assessment	Line-by-line verification
Lowest	Organizational judgment: politics, team dynamics, legal	Retain entirely; do not delegate

its contribution depends on organizational context. The resulting guidance artifact offers three levels of recommendation: directive (strong panel consensus), contextual (moderate agreement, general recommendations), and adaptive (high variability, requires organizational calibration).

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About

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The front cover is created by Thiago A. de Faria.

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Declaration of Authorship

I, Thiago A. de Faria, declare that this thesis titled, 'From GenAI Practice Vacuum to Validated Guidance' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at Antwerp Management School.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at Antwerp Management School or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
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- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: ________________________

Date: ________________________

'Historically, whenever a new AI paradigm has emerged, some AI researchers and many AI businesspeople have claimed "that's it, we found the trick! Within 10 years we'll have machines that are smarter than humans." But they have been consistently over-optimistic. It's always harder than we think.'

Yann LeCun (2025)

'A lot of what I'm spending my brain cycles on is machine learning ways to address safety: particularly how to design AI systems which eventually might be smarter than us, but that will behave well and not harm people.'

Yoshua Bengio (2024)

'People are sufficiently clear and correct about the reasons for local decision making — they know what information is available and how that information is used in deciding on action. But people often do not understand correctly what overall behavior will result from the complex interconnections of known local actions.'

Jay W. Forrester (1995)

'A leader's job is to understand his people, understand their differences; optimize their interactions, their educations, their experiences.'

Edward Deming (1991)

'The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point.'

Claude Shannon (1948)

ANTWERP MANAGEMENT SCHOOL

Abstract

Faculty Enterprise Engineering
Executive Master Enterprise IT Architecture

Master of Enterprise IT Architecture (MSc)

From GenAI Practice Vacuum to Validated Guidance

by THIAGO A. DE FARIA

Generative AI (GenAI) is rapidly integrating into professional knowledge work and software development. Enterprise Architecture (EA) practitioners are using GenAI to analyze systems, draft documentation, and develop Reference Architectures (RAs). Yet no empirical research documents these emerging practices, and no shared guidance exists.

This study uses distributed cognition as its integrating lens to examine how cognitive work redistributes when GenAI enters RA development. We review the literature on EA, RAs, GenAI capabilities, and distributed cognition to build a conceptual model. We then conduct a three-round Design-Oriented Delphi study with EA practitioners to validate emerging practices and produce a guidance artifact.

The central finding is that GenAI acts as a new cognitive partner in the distributed system of EA practice. Practitioners negotiate its role through use: delegating information compression and first drafts, while retaining architectural judgment and stakeholder decisions. The practitioner role shifts from content creator to output curator. The resulting guidance artifact systematizes these practices into directive, contextual, and adaptive recommendations.

Dedicated to Isabela, the love of my life,
and to our children Nina and Liam, who fill it with meaning

Acknowledgments

I would like to thank my family (Isabela, Nina and Liam) for their support during these almost two years. Travelling to Antwerp meant being away from little kids and leaving all the extra work to Isabela. Knowing that I had nothing to worry about at home is a priceless gift. Furthermore, I would like to thank the following people.

Antwerp Management School

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The Delphi Study Participants

A remarkably diverse group in nationalities, work experience and job titles helped me enormously during the Delphi Study. Due to anonymity I cannot reveal their names, but they know who they are. The extensive discussions, arguments and counter-arguments shaped this research in ways I could not have anticipated. In such a fast-paced subject as the adoption of GenAI tools in the tech world, you all recognized the unknowns and showed the willingness to adapt and revise your opinions. Thank you!

All the AI Builders Out There

We live in unprecedented times. I would like to thank the practitioners and builders out there who are pushing software development to new standards and heights: open-source coders releasing tools that are changing the world, and researchers advancing new methodologies, AI safety and ethics. I have been working in the IT industry for 20 years, and what has happened in the last three years displays the disruptive (sorry for using this word) power that technology has. It is up to us, builders, leaders and researchers, to discuss, iterate and ask for help. The consequences of AI pervasiveness, the ethical discussions, the impact on society and on people's practice and identity cannot be forgotten. I am glad the majority of AI builders out there are building in the open.

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List of Abbreviations

AMS	Antwerp Management School
APA	American Psychological Association
API	Application Programming Interface
CLT	Cognitive Load Theory
DCog	Distributed Cognition
EA	Enterprise Architecture
GenAI	Generative Artificial Intelligence
GSS	Group Support System
LLM	Large Language Model
MRQ	Main Research Question
RA	Reference Architecture
sRQ	Sub-Research Question
TOGAF	The Open Group Architecture Framework

1 Introduction

This section establishes the research problem and derives the research questions that guide this study. We begin by situating Generative AI (GenAI) in professional practice, then examine its implications for Enterprise Architecture (EA) and Reference Architecture (RA) development. Two gaps emerge: a practice gap, where no empirical research documents how EA practitioners integrate GenAI into RA development, and a guidance vacuum, where practitioners experiment in isolation without shared frameworks.

The *Theoretical Background* (Section 2) provides the theoretical foundation on EA, RAs, GenAI capabilities, and distributed cognition. The *Conceptual Model* (Section 3) presents the conceptual model that frames the investigation. The *Methodology* (Section 4) explains the Design-Oriented Delphi methodology used to answer the research questions. The following subsections detail this investigation:

1.1 The emergence of Generative AI in Professional Practice

Since the public release of ChatGPT in November 2022, Generative AI (GenAI) has become a routine part of professional work (Brynjolfsson et al., 2025). ChatGPT reached 100 million users within two months (Hu, 2023). By 2025, 84% of developers use AI coding tools, up from 76% one year earlier (Stack Overflow, 2024, 2025), and 75% of knowledge workers across 31 countries report using AI regularly (Microsoft, 2025). Yet adoption has outpaced preparation: 83% of workers are self-taught in AI, and only 52% of senior leaders have deployed formal AI training (EY, 2025).

The infrastructure behind this adoption is large and growing. In 2025, the major US hyperscalers (Amazon, Microsoft, Google, and Meta) are collectively investing approximately $320 billion in AI infrastructure, more than 1% of US GDP (Understanding AI, 2025). Microsoft alone has committed $80 billion to data center construction in fiscal year 2025 (CNBC, 2025). The International Energy Agency (2025) projects that global electricity consumption from data centers will more than double by 2030, reaching 945 TWh (equivalent to Japan's entire current consumption). In the United States, data centers are forecast to account for almost half of electricity demand growth between 2024 and 2030 (International Energy Agency, 2025). McKinsey & Company (2025) estimates the total cost to scale AI data center infrastructure at $7 trillion. These investments have multi-decade lifespans. The practical question is not whether GenAI will persist, but how practitioners should adapt their work to it.

1.2 Reference Architecture Development

Enterprise Architecture (EA) is a discipline that aligns organizational strategy, processes, and IT infrastructure (Lankhorst, 2017). EA practitioners analyze, design, and govern how organizations structure their technology ecosystems (The Open Group, 2022). Among their outputs are Reference Architectures (RAs): documented patterns and templates for system implementations (Cloutier et al., 2010). RAs enable reuse across projects, reduce design risk, promote interoperability, and serve as knowledge repositories for stakeholder communication (Angelov et al., 2012; Cloutier et al., 2010).

In practice, RA development involves more than formal EA titles. Kotusev (2018) showed that EA artifacts are built collaboratively by architects, business executives, analysts, and developers. Solution architects, platform engineers, and senior developers routinely shape these artifacts through daily work. Developing RAs requires analyzing existing systems, identifying patterns, engaging stakeholders, and synthesizing technical and business considerations.

GenAI is already entering this work. Developer tools powered by GenAI are shaping architectural patterns from the bottom up, through code generation, design suggestions, and automated documentation, often before formal EA governance becomes aware (Peng et al., 2023; Tabarsi et al., 2025). This makes understanding how practitioners use GenAI in RA development urgent.

1.3 The Practice Gap

No empirical research documents how EA practitioners integrate GenAI into RA development[1]. Research in adjacent fields (software engineering, management consulting, general knowledge work) shows that GenAI changes professional practice (Dell'Acqua et al., 2023; Noy & Zhang, 2023; Peng et al., 2023), but EA remains unstudied (Banh et al., 2025; Ettinger, 2025).

This gap matters for three reasons. First, EA work involves both technical and social dimensions. Practitioners navigate organizational politics, synthesize diverse perspectives, and create artifacts that must resonate with multiple stakeholders. How GenAI tools, primarily trained on technical content, interact with this sociotechnical reality is unknown. Second, RA development is knowledge-intensive. It requires capturing, synthesizing, and codifying architectural knowledge. GenAI's capabilities for pattern recognition and content generation could change how this knowledge work is done, but no evidence documents whether or how. Third, research in other domains shows that professionals restructure their work practices around AI rather than simply using it as a faster tool (Dell'Acqua et al., 2023; Noy & Zhang, 2023; Tankelevitch et al., 2024). Whether EA practitioners are experiencing similar changes is unexamined.

1.4 The Guidance Vacuum

Practitioners who experiment with GenAI in EA work do so without shared vocabulary, evaluation criteria, or validated guidance. Specifically, they lack:

1. Evidence of which practices work for GenAI use in EA (Banh et al., 2025).
2. Frameworks for judging when GenAI helps versus hinders architectural activities (Esposito et al., 2026).
3. Guidance on managing the risks of human-AI collaboration in EA, particularly overreliance (Dell'Acqua et al., 2023).

[1] EBSCO host advanced search conducted on June 8, 2025, resulted in 0 publications using the query: ((enterprise architecture OR ea practice) AND (gen ai OR generative artificial intelligence OR llm OR large language models)) AND SO (MIS Quarterly Executive OR Information Systems Research OR Journal of the Association for Information Systems OR European Journal of Information Systems) Filters applied: peer-reviewed articles, publication date from January 1, 2020, to June 8, 2025

Without shared guidance, each practitioner reinvents the wheel (Feldman & Orlikowski, 2011). Emerging practices cannot consolidate into reliable routines, and organizations risk both missed opportunities and quality problems (Dell'Acqua et al., 2023).

1.5 Research Opportunity

In our view, these gaps call for an investigation into how EA practitioners develop new practices as they integrate GenAI into RA development. This is not a tool-adoption study; Section 3.4 develops this argument further. Research in adjacent domains shows that professionals reshape their work around AI capabilities rather than bolting AI onto existing workflows (Dell'Acqua et al., 2023; Noy & Zhang, 2023).

We focus on RA development because it is a central EA activity (Cloutier et al., 2010; The Open Group, 2022) that spans the full range of architectural work: analysis, synthesis, and documentation. If GenAI changes how RAs are developed, the effects will extend to governance, implementation, and knowledge management. Beyond documenting current practices, the aim is to develop practical guidance that supports EA practitioners in navigating this change.

In sum, GenAI is a structural shift in professional knowledge work. RA development is knowledge-intensive sociotechnical work for which no empirical guidance exists. Developer-level GenAI use is already shaping architectural artifacts before formal EA governance becomes aware. The following subsections translate this problem into research questions.

1.6 Deriving the Research Questions

The introduction identified two gaps. A *practice gap*: no empirical research documents how EA practitioners integrate GenAI into RA development. And a *guidance vacuum*: practitioners experiment in isolation, without shared frameworks or vocabulary. These gaps call for research that both documents emerging practices and produces actionable guidance.

1.7 Main Research Question

> *What practices do EA practitioners adopt when using GenAI for RA development, and how can these be systematized into guidance?*

The first part of this question targets practice discovery. The second targets guidance development. This reflects a practice-based perspective: technology integration is not a one-time event but an ongoing process shaped by how people actually use tools in context (Feldman & Orlikowski, 2011; Orlikowski, 2000). GenAI and EA practice shape one another. Practitioners adapt their workflows to GenAI, while GenAI outputs depend on practitioner expertise and organizational context (Orlikowski, 2007). Answering this question requires a methodology that captures both current practices and the ambition to produce practical guidance.

1.8 Sub-Questions

Four sub-questions decompose the main RQ into a progression from description through analysis and interpretation to design:

sRQ1: What GenAI practices are EA practitioners currently using when developing RAs?

sRQ2: What changes to EA practice are EA practitioners experiencing when RAs are developed using GenAI?

sRQ3: How do EA practitioners make sense of impacts and trade-offs when adopting GenAI for RA development?

sRQ4: How should effective GenAI practices for RA development be systematized into guidance for EA practitioners?

Throughout these questions, "EA practitioners" refers broadly to all professionals contributing to RA development, as formally defined in Section 2.

1.9 Sub-Question Progression

The four sub-questions follow a deliberate sequence. sRQ1 is *descriptive*: it maps which GenAI practices EA practitioners currently use. sRQ2 is *analytical*: it examines how these practices change existing EA work. sRQ3 is *interpretive*: it surfaces how practitioners weigh the impacts and trade-offs they experience.

sRQ4 differs from the first three. Where sRQ1 through sRQ3 investigate what *is*, sRQ4 asks what *should be*. It synthesizes empirical findings into a guidance artifact that practitioners can use (Gregor & Hevner, 2013). This shift from empirical inquiry to artifact construction is characteristic of the Design-Oriented Delphi methodology adopted in this study (Section 4).

2 Theoretical Background

Introduction

This section defines the five constructs that ground the study: Enterprise Architecture (EA), EA practitioners, EA practice, Reference Architectures (RAs), and Generative AI (GenAI). It then introduces distributed cognition as the theoretical lens that connects them. The theoretical background from this section is used in the construction of the conceptual model in the *Conceptual Model* (Section 3). The following subsections detail these foundations:

2.1 Enterprise Architecture

Enterprise Architecture has roots in IBM's Business Systems Planning methodology of the 1960s (Kotusev, 2018) and emerged as a distinct discipline with Zachman (1987)'s framework. After nearly four decades, the field still lacks a shared definition. Saint-Louis (2019) found that EA means different things to different people, with definitions varying across theoretical lenses and organizational contexts.

The Open Group (2022) defines EA as the practice of analyzing, designing, planning, and implementing enterprise-level analysis in service of business strategy. Lankhorst (2017) presents EA as an integrated set of principles, methods, and models for designing an enterprise's organisational structure, business processes, information systems, and infrastructure. The first definition treats EA as an active practice; the second treats it as both process and product.

Saint-Louis (2019) identified a useful distinction: EA as a noun versus EA as a verb. As a noun, EA refers to deliverables, models, and artifacts. As a verb, it refers to the practice and process of architecting. This distinction matters for this study because GenAI may affect both what practitioners produce and how they produce it.

The lack of definitional consensus has practical consequences. Kotusev (2018) notes that EA frameworks offer vague, inconsistent, and conflicting recommendations, creating problems for organizations trying to implement EA. Some view EA as primarily IT-focused; others emphasize business-IT alignment or enterprise-wide transformation (Gartner, 2008; Ross et al., 2006).

In our view, EA is best understood as the sociotechnical practice of analyzing, designing, and governing an enterprise's structures, processes, and technology to enable strategic alignment. This working definition bridges the noun and verb perspectives: EA produces tangible artifacts (models, standards, principles) through an ongoing practice of stakeholder negotiation, technical design, and organizational adaptation (Lankhorst, 2017; Saint-Louis, 2019; The Open Group, 2022).

2.1.1 Historical Evolution

EA's origins are commonly traced to Zachman (1987)'s "A Framework for Information Systems Architecture." This work introduced multiple perspectives and abstraction levels for describing enterprises. Zachman established the principle that a single complex object can be described in multiple ways, depending on the purpose and type of description required (Zachman, 1987). Early EA was largely understood through an engineering metaphor, with practitioners viewing themselves as technical specialists designing optimal

IT solutions (Spewak & Hill, 1993).

The discipline shifted in the 2000s. Ross et al. (2006) repositioned EA as the organizing logic that aligns core business processes and IT infrastructure with a firm's integration and standardization requirements (Ross et al., 2006). This moved EA from documentation toward strategic capability. The Open Group's TOGAF framework, first published in 1995, gained prominence through successive revisions (The Open Group, 2022).

More recently, Kotusev (2018) challenged many assumptions about EA in practice. He argues that the current conceptualization rests on anecdotal stories and wishful thinking rather than empirical evidence. Through case study research, he showed that successful EA practices diverge from mainstream prescriptions. Rather than following a linear path toward maturity, EA practices adapt to organizational contexts, with successful implementations often bearing little resemblance to framework prescriptions.

Saint-Louis (2019) similarly notes that despite decades of development, EA still lacks common understanding. EA continues to evolve, with practitioners and researchers still working out its boundaries and methods.

2.1.2 EA as Sociotechnical Practice

EA is not purely technical work. Saint-Louis (2019) identified three contexts within which EA operates, each combining technical and social dimensions:

1. Technological: IT infrastructure, systems integration, and technical optimization
2. Socio-Technological: stakeholder participation, communication, and collaborative decision-making
3. Eco-Technological: ecosystem relationships, compliance requirements, and external standards

EA practice requires integrating both technical and social considerations in specific organizational settings. Gregor et al. (2007) demonstrated through case studies that effective EA development requires stakeholder participation, with solutions achieved through communication, negotiation, and collaboration. Lankhorst (2017) similarly characterizes EA as a holistic approach to organisational design, where multiple domains (organization, information, systems, products, processes, and applications) intersect. This sociotechnical character is why GenAI integration in EA is not simply a tool-adoption question.

2.2 EA Practitioners and EA Practice

2.2.1 EA Practitioners: Roles and Evolution

EA practitioners are the professionals who develop, maintain, and govern enterprise architectures (Kotusev, 2018; The Open Group, 2022). Kotusev (2018) notes that EA artifacts are developed not only by dedicated architects but also by business executives, business analysts, project managers, and software developers. EA practice therefore extends beyond formal "Enterprise Architect" job titles.

Saint-Louis (2019) identified three practitioner archetypes:

1. **Specialists**: technical experts focused on selecting optimal solutions for organizational needs (Saint-Louis, 2019).
2. **Integrators**: practitioners who prioritize stakeholder participation and motivation in the decision-making process (Saint-Louis, 2019). They bridge technical and business domains through negotiation.
3. **Facilitators**: professionals who view IT implementation and stakeholder engagement as inseparable from broader organizational adaptation (Saint-Louis, 2019).

In practice, EA practitioners shift between these roles depending on the situation. Land et al. (2009) observe that effective architects need competencies spanning technical expertise, communication, leadership, and political acumen. van Steenbergen (2011) shows that EA effectiveness depends on maturity across multiple dimensions, not technical capability alone.

Our position is that "EA practitioners" in this research encompasses all professionals who contribute to the development and governance of architectural artifacts: enterprise architects, solution architects, platform engineers, data architects, and senior software engineers who shape RAs through their daily work. Kotusev (2018)'s finding of distributed authorship supports this broader scope.

2.2.2 EA Practice: Beyond Frameworks

We understand EA practice, following Feldman and Orlikowski (2011), as the recurrent patterns of action through which practitioners accomplish architectural work. These practices are sociomaterial accomplishments (Orlikowski, 2007): they involve both human actors and technological artifacts.

Kotusev (2018)'s empirical research revealed four disconnects between formal EA frameworks and actual practice. Ross et al. (2006), Fonstad, N. O. and Robertson, D. (2006), and Ahlemann (2012) corroborate these findings:

1. **Incremental Development**: EA artifacts evolve gradually through organizational learning, not all at once (Kotusev, 2018).
2. **Distributed Authorship**: Business executives are actively involved, sometimes leaving enterprise architects in a secondary facilitator role (Kotusev, 2018; Ross et al., 2006).
3. **Broad Application**: EA applications extend beyond strategic initiatives to address local requirements, unforeseen demands, and unplanned deviations (Fonstad, N. O. & Robertson, D., 2006).
4. **Process Integration**: Successful EA integrates with strategic, project, and operations management (Ahlemann, 2012).

The point is that EA work happens through dynamic interactions among practitioners, tools, and organizational structures. As Feldman and Orlikowski (2011) argue, understanding practice means examining how activities are generated and operate within different contexts and over time (Feldman & Orlikowski, 2011, p. 1241).

2.3 Reference Architectures and Their Lifecycle

Cloutier et al. (2010) define a Reference Architecture (RA) as an authoritative knowledge source for a specific domain that guides and constrains how concrete architectures and solutions are instantiated. Angelov et al. (2012) complement this by characterizing RAs as generic architectures for a class of information systems, designed to provide standardization, interoperability, and reuse. RAs serve four functions: they capture proven architectural patterns, control complexity through abstraction, promote common understanding among stakeholders, and reduce risk through reuse of validated designs (Cloutier et al., 2010).

Cloutier et al. (2010) describe how RAs are built from proven architectures of past and existing products yet provide guidance for future implementations. RAs must therefore evolve with technological change and organizational learning. However, the literature does not present a unified view of the RA lifecycle. The following subsections review five perspectives, then synthesize them into an operational definition and sensitizing lifecycle model. This model serves as a structuring device for the Delphi study: Round 1 participants position their practices within the lifecycle phases, providing empirical validation of its utility (Section 4).

2.3.1 Perspective 1: Continuous Development Cycle

Cloutier et al. (2010) present a development cycle showing the relationship between existing architectures, RA patterns, and new implementations. In their view, RAs should evolve continuously because organizational needs, technologies, and standards change. This requires active maintenance and refactoring (Cloutier et al., 2010). They stress that the Chief Architect (business plus technical) owns the RA and that business sponsorship is a prerequisite.

2.3.2 Perspective 2: Evolution and Transformation

Angelov et al. (2012) focus on how RAs evolve over time. Using their classification framework (which categorizes RAs by context, goal, and scope), they note that an RA may shift categories as its environment changes or its organizational goal evolves. Examples of evolution paths include:

- A preliminary RA transforming into a classical architecture when the domain matures
- Evolution from facilitation to standardization as organizational needs change
- Expansion from single organization scope to multiple organizations

To remain usable, an RA must be adapted to reflect such changes (Angelov et al., 2012).

2.3.3 Perspective 3: Creation and Application Context

Creating RAs involves analyzing existing systems, identifying commonalities and variations, and abstracting from implementation specifics (Angelov et al., 2012). Application requires careful attention to context, since the generic nature of RAs leads to less defined design and application boundaries. Cloutier et al. (2010) describe the challenge of making RAs accessible to multiple stakeholders (engineers, business managers, customers) while keeping them generic enough for broad reuse yet concrete enough to be actionable. This tension between abstraction and concreteness cannot be resolved through technical means alone. It requires ongoing negotiation with the stakeholders who must interpret and apply the guidance.

2.3.4 Perspective 4: Stakeholder Engagement and Validation

The literature emphasizes intensive stakeholder involvement throughout the RA's existence. Ross, J. W. (2004) found that developing an architectural vision required approximately sixty meetings of the executive team, illustrating the collaboration that architectural work demands. Angelov et al. (2012) note that RA quality can be measured by its success in fulfilling its goals, which requires ongoing validation through implementation and feedback.

2.3.5 Perspective 5: Ongoing Refinement

Rather than following a linear lifecycle, the literature suggests RAs exist in a state of continuous adaptation. Cloutier et al. (2010) explain that RAs must be actively managed to ensure they are updated based on evolving standards, emerging technology, changing stakeholder requirements, and maturing business needs. The RA lifecycle is an ongoing process of refinement, not a sequence of phases.

2.3.6 Operational Definition for This Research

The five perspectives can be summarized as follows:

1. **Continuous Development Cycle**: RAs require active maintenance, not one-time creation (Cloutier et al., 2010).
2. **Evolution and Transformation**: RAs shift across categories as technology, experience, and goals change (Angelov et al., 2012).
3. **Creation and Application Context**: RA design must balance abstraction for reuse with concreteness for implementation (Angelov et al., 2012; Cloutier et al., 2010).
4. **Stakeholder Engagement**: Architectural work demands extensive multi-stakeholder collaboration (Angelov et al., 2012; Ross, J. W., 2004).
5. **Ongoing Refinement**: RAs exist in continuous adaptation driven by implementation experience (Cloutier et al., 2010).

RA work extends well beyond initial creation. For this study, Reference Architecture encompasses templates, standards, patterns, governance documents, blueprints, and domain models used as reference for downstream architectural design, regardless of formality level. Project-specific solution architectures fall outside this scope, unless they are reused as reference for other projects, at which point they function as RAs.

This broad scope reflects how practitioners engage with RAs in practice. RA work encompasses creating, reading, absorbing, contextualizing, validating, communicating, governing, and evolving architectural artifacts. Cloutier et al. (2010) describe a continuous development cycle in which RAs require active organizational management. Angelov et al. (2012) note that RAs must evolve to reflect changes in technology, experience, and goals.

2.3.7 Sensitizing Lifecycle Model

The Delphi study requires participants to position their GenAI practices within the RA lifecycle. However, the five perspectives describe lifecycle *characteristics* (continuous, iterative, stakeholder-intensive) without converging on named phases. To provide a shared vocabulary for data collection, we synthesize an eight-phase lifecycle model:

1. Stakeholder Analysis & Requirements Gathering
2. Current State Analysis
3. Pattern Identification & Abstraction
4. Target State Design
5. Component Specification
6. Documentation & Communication
7. Validation & Review
8. Evolution & Maintenance

Each phase maps to the perspectives discussed above. Phases 1–2 correspond to requirements and context analysis (Angelov et al., 2012). Phases 3–5 reflect the creation and abstraction activities in the development cycle (Angelov et al., 2012; Cloutier et al., 2010). Phase 6 addresses the multi-stakeholder accessibility that Cloutier et al. (2010) identify as essential. Phase 7 captures the quality validation that Angelov et al. (2012) emphasize. Phase 8 embodies the ongoing refinement both sources describe.

This model serves as a *sensitizing device*, not a prescriptive sequence. Practitioners may enter at any phase, iterate, skip phases, or work across multiple phases simultaneously. Its purpose is to provide a common vocabulary for participants to position their GenAI practices within the RA lifecycle. The model is used as the data collection instrument in Section 4, where Round 1 workbook participants map their practices to these eight phases.

2.4 Generative AI: Emergence and Capabilities

Artificial intelligence research has progressed through several waves: from rule-based expert systems in the 1980s, through machine learning and deep learning approaches that required curated training data, to the current generation of large-scale models that learn from broad corpora and generate novel outputs (Kanbach et al., 2024). Generative AI (GenAI) refers to this latest class of AI systems, capable of generating new content (text, images, audio, code, and video) based on patterns learned from training data (Kanbach et al., 2024).

Vaswani et al. (2017) introduced the transformer architecture that underlies current GenAI systems, enabling attention mechanisms that capture global dependencies without relying on recurrence or convolutions. Large Language Models (LLMs) are specific GenAI implementations focused on natural language. These models are trained on vast text corpora using self-supervised learning (Brown et al., 2020). The key innovation is not information retrieval but the ability to generate coherent, contextually appropriate text by combining learned patterns in novel ways (Bubeck et al., 2023).

2.4.1 Rapid Adoption

As documented in Section 1, GenAI adoption has been rapid and broad. The public release of ChatGPT in November 2022 reached 100 million users within two months (Hu, 2023), and by 2025 the majority of developers and knowledge workers use AI tools regularly. This adoption has occurred despite limited understanding of best practices, creating what Dell'Acqua et al. (2023) describe as a "jagged technological frontier" where capabilities are unevenly distributed across task types.

2.4.2 Core Capabilities Relevant to EA Practice

GenAI systems exhibit several capabilities relevant to architectural work:

1. **Pattern Recognition and Synthesis**: GenAI can identify patterns in unstructured content (text, diagrams, code) and synthesize novel outputs that combine them (Brynjolfsson et al., 2025). For EA practitioners, this enables rapid analysis of existing architectures.
2. **Context-Aware Generation**: Modern LLMs maintain context windows of hundreds of thousands of tokens, allowing extended interactions and iterative refinement of architectural artifacts (Banh et al., 2025; Kanbach et al., 2024).
3. **Multi-Modal Reasoning**: Recent models process and generate text, code, images, and structured data (Kanbach et al., 2024), supporting the diverse representational needs of architectural work.
4. **Knowledge Integration**: GenAI can combine information from multiple domains, bridging technical and business perspectives (Banh et al., 2025).

However, GenAI also introduces challenges: hallucination (generating plausible but incorrect information) (Y. Zhang et al., 2023), lack of genuine understanding, and potential for bias inherited from training data (Bender et al., 2021).

2.4.3 GenAI Integration in Professional Practice

Emerging research reveals patterns of GenAI integration relevant to EA practice. Noy and Zhang (2023) found that professionals using ChatGPT for writing tasks decreased inequality between workers, with less skilled workers benefiting more while overall productivity increased by 37%.

Dell'Acqua et al. (2023) identified two patterns of AI use among consultants. Centaurs strategically divide tasks between themselves and AI. Cyborgs integrate AI into their workflow for all tasks. Their research showed that professionals develop contextual understanding about when AI helps versus hinders.

Tankelevitch et al. (2024) documented new metacognitive demands when professionals work with AI:

1. Evaluating AI-generated content for accuracy and relevance
2. Managing the balance between efficiency and critical thinking
3. Developing strategies for effective prompting
4. Maintaining domain expertise despite AI assistance

H.-P. H. Lee et al. (2025) surveyed 319 knowledge workers using GenAI tools at work and found that GenAI shifts critical-thinking effort from material production to AI oversight, with workers reporting reduced effort for information gathering but greater effort for verifying and integrating AI outputs. Taken together, these findings show that GenAI does not merely automate tasks. It redistributes cognitive work between human and machine, which requires a theoretical framework to analyze.

2.5 Distributed Cognition as Integrating Lens

2.5.1 Distributed Cognition Theory

Distributed cognition (Hutchins, 1995) provides a framework for understanding how cognitive processes spread across the members of a social group, the tools they use, and the environment in which activity takes place. Rather than treating cognition as something that happens only inside individual brains, this perspective examines how cognitive systems accomplish tasks through the propagation of representational states across media.

Hutchins (1995) studied navigation teams and showed that the computational work of navigation is accomplished through physical manipulation of representations distributed across team members, instruments, charts, and procedures. The cognitive system includes all these elements working together. No individual component could accomplish the task alone.

Hollan et al. (2000) extended this framework to human-computer interaction, identifying three types of cognitive distribution:

1. Cognitive processes distributed across group members
2. Cognitive processes involving coordination between internal and external structures
3. Processes distributed through time, with products of earlier events transforming later events

2.5.2 Application to EA-GenAI Integration

Hutchins (1995) canonical study of ship navigation analyzed the cognitive system as distributed across four substrate categories: team members, instruments, external representations, and the procedures coordinating their use. We adopt this same decomposition for the EA-GenAI system. Each element below maps to a substrate category recognised in the distributed cognition literature (Hollan et al., 2000; Perry, M., 2003).

In GenAI-augmented architectural work, cognitive resources are distributed across four elements:

1. **EA Practitioners**: domain knowledge, organizational understanding, stakeholder awareness, and evaluative judgment (Perry, M., 2003)
2. **GenAI Systems**: pattern recognition, content generation, rapid synthesis, and access to encoded knowledge (Brynjolfsson et al., 2025)
3. **Organizational Artifacts**: external memory and constraint structures (J. Zhang & Norman, 1994)
4. **Collaborative Tools**: coordination among human and artificial agents (Brynjolfsson et al., 2025)

A clarification: the three types of distribution from Hollan et al. (2000) describe *how* cognition distributes (across people, between internal and external structures, and through time). The four elements above identify *across what* cognitive resources are distributed in the EA-GenAI system. This decomposition serves as the analytical lens through which Sections 5 and 8 examine how introducing GenAI restructures the cognitive system for RA development. The Delphi study empirically tests whether this theoretical mapping holds.

As Hutchins (1995) argues, the functional properties of a cognitive system cannot be reduced to the properties of individual agents. The distributed system may produce capabilities neither human nor AI possesses alone. In our view, this explains why simple tool-adoption perspectives miss what is happening when GenAI enters EA work. The unit of analysis must be the larger system (Hollan et al., 2000), not the individual practitioner.

2.6 Core Constructs for This Study

This study examines the interaction of five constructs:

1. **Generative AI (GenAI)**: the technological capabilities introduced into the EA ecosystem, serving as a cognitive partner within a distributed system (Hutchins, 1995; Vaswani et al., 2017).
2. **EA Practitioners**: professionals who operate as specialists, integrators, or facilitators (Saint-Louis, 2019), spanning roles from enterprise architect to platform engineer.
3. **Reference Architecture**: the focal artifact for studying practice evolution, capturing accumulated architectural knowledge (Cloutier et al., 2010).
4. **EA Practice**: the recurrent patterns of action (Feldman & Orlikowski, 2011) within which architectural work is accomplished.

5. **Guidance (Research Output)**: the prescriptive output that synthesizes findings into actionable recommendations, validated through panel deliberation.

These five constructs and their interrelationships form the basis of the conceptual model in the following section.

3 Conceptual Model

Introduction

The *Theoretical Background* (Section 2) defined five constructs (EA, EA practitioners, EA practice, RAs, and GenAI) and introduced distributed cognition as the integrating lens. This section synthesizes those elements into a conceptual model that frames the investigation and maps each construct relationship to a sub-research question. The following subsections detail the model:

3.1 Model Development

Four theoretical streams inform the model. Distributed cognition (Hollan et al., 2000; Hutchins, 1995) treats architectural work as a cognitive system that spans human and artificial agents. When EA practitioners use GenAI, the two form a functional system (Hutchins, 1995): neither holds the full capability alone, and the division of labour changes how the work gets done.

Practice theory (Feldman & Orlikowski, 2011) adds that EA work consists of recurrent patterns of action. These patterns develop through actual use, not from technology specifications alone (Orlikowski, 2007).

The sociotechnical character of EA (Lankhorst, 2017; Saint-Louis, 2019) means that architectural work sits inside organizational contexts. Practitioners must navigate stakeholder perspectives, governance constraints, and external pressures across technological and socio-technological layers (Saint-Louis, 2019).

The literature on Reference Architecture (Angelov et al., 2012; Cloutier et al., 2010) shows that RA development involves continuous cycles of abstraction, synthesis, validation, and evolution. RAs function as boundary objects: they must be understandable to multiple stakeholders while capturing reusable architectural knowledge (Cloutier et al., 2010).

3.2 The Conceptual Framework

Figure 1 presents the conceptual framework. Four elements connect through specific relationships that form a cycle of practice evolution. Guidance sits at the centre as a synthesizing output that both emerges from and feeds back into this cycle. Each relationship corresponds to a sub-research question (sRQ1–sRQ4), as defined in Section 1.8.

Elements and relationships are analysed across individual-collective, descriptive-prescriptive, and temporal dimensions (see Section 3.2.1 for tractability within the Delphi design). The five core constructs (GenAI, EA Practitioners, Reference Architecture, EA Practice, and Guidance) were defined in *Core Constructs for This Study* (Section 2.6). The following subsections detail the analytical dimensions and the specific relationships between these constructs.

3.2.1 Framework Dimensions

Beyond the structural relationships, three analytical dimensions shape how GenAI integration plays out in practice. Each dimension varies in how directly the Delphi design can capture it.

1. **Individual-Collective**: Distributed cognition theory (Hutchins, 1995) holds that

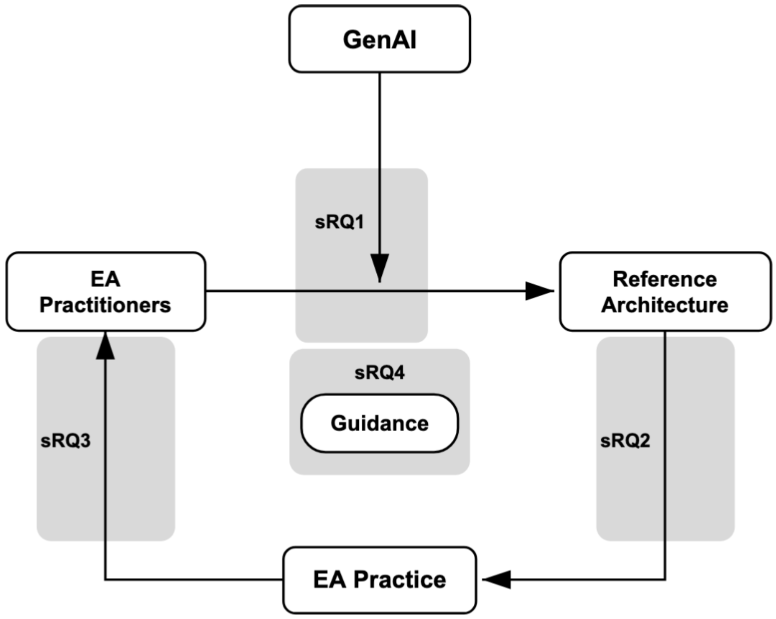

Figure 1. Analytical Framework for Investigating EA-GenAI Integration.

practices develop at both individual and collective levels. Practitioners build personal approaches while contributing to shared organizational patterns. The Delphi captures individual practices through self-report (Round 1) and surfaces collective patterns through group deliberation (Rounds 2–3). It does not, however, collect organizational-level data; collective dynamics are inferred from practitioner accounts.

2. **Descriptive-Prescriptive**: The framework separates what is happening now (descriptive) from what should happen (prescriptive). The descriptive side draws on empirical data from all three rounds. The prescriptive side is the researcher's synthesis of that input into a guidance artifact, validated through panel consensus (sRQ4) rather than field implementation.
3. **Temporal**: Practices move from initial experimentation through pattern emergence to stabilization. Practitioners and organizations sit at different points on this curve. Within the study, temporal dynamics appear indirectly: practitioners give retrospective accounts, and the three rounds span roughly four months. Longitudinal observation falls outside the study's scope but remains an important avenue for future research.

These dimensions are not separate layers; they cut across all elements and relationships. The Delphi captures the individual-descriptive intersection most directly, while collective and temporal dynamics are approximated through panel deliberation and retrospective accounts.

3.3 Framework Relationships and Manifestations

Each sub-research question examines a specific relationship within the framework. Understanding each relationship clarifies what the research must capture and what the methodology must support (Section 4).

The subsections below detail each relationship. For sRQ1–sRQ3, the analysis identifies observable manifestations that inform data collection. For sRQ4, it addresses how insights from the preceding relationships feed into actionable guidance. Table 3 summarizes the mapping.

Table 3. Mapping of Sub-Questions to Framework Relationships

Sub-Question	**Framework Relationship**	**Research Focus**	**Methodological Requirement**
sRQ1	GenAI → Reference Architecture (via EA Practitioners)	Current practices	Discovery of emerging patterns
sRQ2	Reference Architecture → EA Practice	Practice changes	Capturing evolution effects
sRQ3	EA Practice → EA Practitioners	Sense-making	Understanding interpretations
sRQ4	All relationships → Guidance	Systematization	Consensus building

3.3.1 sRQ1: EA Practitioner-AI Collaboration Patterns

"What GenAI practices are EA practitioners currently using when developing RAs?"

This relationship examines how practitioners divide work between themselves and GenAI when building RAs. Distributed cognition theory (Hutchins, 1995) frames this as a redistribution of cognitive labour across human and artificial agents.

Observable manifestations include: which RA activities practitioners delegate to GenAI versus reserve for human judgment; whether GenAI is consulted continuously or at specific lifecycle phases; how sophisticated prompt strategies have become; and which tools or models practitioners prefer for architectural work.

These patterns develop at both individual and collective levels. Individual experimentation feeds into shared organizational learning, though the pace varies across practitioners and organizations.

3.3.2 sRQ2: Artifact-Driven Practice Evolution

"What changes to EA practice are EA practitioners experiencing when RAs are developed using GenAI?"

This relationship traces how RAs built with GenAI change broader EA practice. Because sRQ2 asks what changes practitioners experience, the framework follows the causal path from GenAI-augmented artifacts back into the practice context. Angelov et al. (2012) describe a similar dynamic: reference architectures influence the application contexts that produced them.

Manifestations include changes in both artifacts and processes. RAs developed with GenAI may show different structural patterns, documentation styles, or architectural approaches. Development cycles may speed up or demand new validation steps. Governance, implementation projects, and knowledge management practices may all shift as organizations encounter AI-augmented artifacts.

As Kotusev (2018) observed, successful EA practices often grow bottom-up rather than top-down. GenAI-augmented RAs can accelerate that kind of organic evolution.

3.3.3 sRQ3: EA Practitioner Sense-Making and Adaptation

"How do EA practitioners make sense of impacts and trade-offs when adopting GenAI for RA development?"

This relationship captures the feedback loop: evolving EA practices shape how practitioners interpret their situation and decide what to do next. Orlikowski and Gash (1994)'s concept of technological frames explains how practitioners update their mental models in response to practice changes.

Manifestations span cognitive and practical levels. Practitioners develop rules of thumb for when GenAI adds value versus when it risks compromising architectural quality. They build validation routines for AI-generated content. Professional identity shifts as traditional expertise meets new orchestration skills. Risk awareness grows as practitioners learn the limits of human-AI collaboration.

This sense-making is collective, not only individual. Interpretations are shared, debated, and refined within practitioner communities. The resulting shared understanding shapes how practitioners approach future GenAI interactions, closing the feedback loop.

3.3.4 sRQ4: Synthesis into Guidance

"How should effective GenAI practices for RA development be systematized into guidance for EA practitioners?"

The fourth sub-question sits at a meta-level: it draws from all three relationships to produce prescriptive recommendations. From sRQ1, guidance captures effective collaboration approaches. From sRQ2, it documents quality criteria and process adaptations. From sRQ3, it incorporates practitioner wisdom about trade-offs and challenges.

Turning descriptive findings into prescriptive guidance requires assessing practice maturity, contextual fit, and compatibility with existing EA frameworks. Because the transformation is researcher-led and validated through panel consensus, the resulting prescriptions reflect collective judgment rather than field-tested outcomes.

As Dell'Acqua et al. (2023) observe, AI capability varies unpredictably across tasks, creating a "jagged technological frontier." The guidance must accommodate this unevenness. Where practitioner consensus is strong, directive recommendations are feasible; where it is not, guidance must remain adaptive. The final artifact form depends on Round 3 findings.

3.4 Framework Dynamics

The theoretical review surfaced three reasons why a simple tool-adoption lens falls short. First, EA work is sociotechnical; it requires negotiation among diverse stakeholders (Saint-Louis, 2019). Second, GenAI redistributes cognitive work rather than automating it (Dell'Acqua et al., 2023). Third, professionals restructure their practices around AI rather than merely using it as a faster tool (Noy & Zhang, 2023).

These observations point to three dynamics that run through the framework:

1. **Distributed cognition**: the capability of the practitioner-GenAI system exceeds what either party contributes alone (Hutchins, 1995). GenAI provides pattern recognition across large knowledge spaces; practitioners provide contextual judgment and stakeholder awareness.
2. **Emergent evolution**: practices arise from situated use (Orlikowski, 2000), not from technology specifications. Initial experiments lead to new RA approaches, which shift practice patterns, which update practitioner understanding, which drives further experimentation.
3. **Multi-level simultaneity**: individual experimentation and collective pattern formation happen in parallel. A single practitioner's GenAI session is both a personal learning event and a contribution to organizational knowledge.

Because the phenomenon keeps evolving, any guidance produced must be designed for periodic revision as collective understanding matures.

4 Methodology

Introduction

The conceptual framework (*Conceptual Model*, Section 3) calls for a method that can tap distributed expertise, detect new practices as they form, and produce both analysis and practical guidance. This section describes how a three-round Design-Oriented Delphi study meets those requirements. Round 1 ran asynchronously (individual discovery workbooks, January 2026); Rounds 2–3 ran as synchronous sessions (February–March 2026) using Group Support System (GSS) technology: software for structured, anonymous group deliberation through parallel input, voting, and real-time aggregation (Bobbert, 2017; Bobbert & Mulder, 2013; Klein et al., 2007). The following subsections detail the research design and implementation:

4.1 Methodological Approach: The Delphi Method

4.1.1 Defining the Delphi Method

The Delphi method is a structured communication technique for enabling effective group deliberation on complex problems where incomplete knowledge exists (Linstone & Turoff, 1975). The method employs iterative rounds of data collection and analysis, with controlled feedback between rounds, to aggregate expert judgment while managing group dynamics (Dalkey & Helmer, 1963).

Classical Delphi characteristics include:

1. **Structured communication**: Information flows through a defined process rather than unstructured discussion.
2. **Iteration with feedback**: Multiple rounds allow participants to refine their views based on collective input.
3. **Controlled feedback**: Researchers synthesize and present group responses systematically.
4. **Expert participation**: Participants are selected based on relevant expertise.
5. **Anonymity**: Participants provide input without direct confrontation, reducing dominance effects.

This study incorporates all five characteristics, as detailed in the research design below.

4.1.2 Justification for an Exploratory-Convergent Design

GenAI integration in EA practice is new enough that imposing predetermined categories would miss novel practices (Okoli & Pawlowski, 2004). The research therefore begins with open discovery before moving to structured convergence (Schmidt, 1997). Synchronous rounds then compress the timeline while maintaining iterative refinement, and combining asynchronous depth with synchronous interaction draws on the strengths of both modes (Gordon & Pease, 2006).

4.2 Research Design: Exploratory-Convergent Design-Oriented Synchronous Delphi

4.2.1 Positioning Within Delphi Typology

The study combines three Delphi orientations. Exploratory Delphi enables open-ended discovery of undocumented practices (Okoli & Pawlowski, 2004). Convergent Delphi provides mechanisms for progressive synthesis toward consensus (Linstone & Turoff, 1975).

Design Delphi focuses on producing actionable artifacts (Skulmoski et al., 2007). Day and Bobeva (2005) argue that Delphi designs should be tailored to research purposes rather than following a single typology; this study does so by sequencing exploration, convergence, and design so that each phase builds on the outputs of the previous one. Each component follows established Delphi principles (iteration, controlled feedback, expert participation), and documented protocols for every phase support replication.

4.2.2 The Emergent Artifact Approach

Building on Delphi scholarship that emphasizes responsiveness to participant input (Linstone & Turoff, 1975), this study adopts a practitioner-grounded emergence approach. The guidance artifact develops progressively from practitioner data rather than being presented as a preliminary instrument in any round. Specifically:

1. **Round 1: No artifact presented**: practitioners document their discovered practices
2. **Between Rounds 1–2**: researcher synthesizes practices into initial patterns
3. **Round 2**: emergent patterns presented for validation
4. **Between Rounds 2–3**: researcher refines artifact based on validation data
5. **Round 3**: practitioners critique a refined draft artifact and provide input for remaining gaps
6. **Post-Round 3**: researcher finalizes guidance artifact

As illustrated in Figure 2, the guidance artifact evolves through iterative rounds, beginning with practitioner-generated data and progressively refined through researcher synthesis and collective validation.

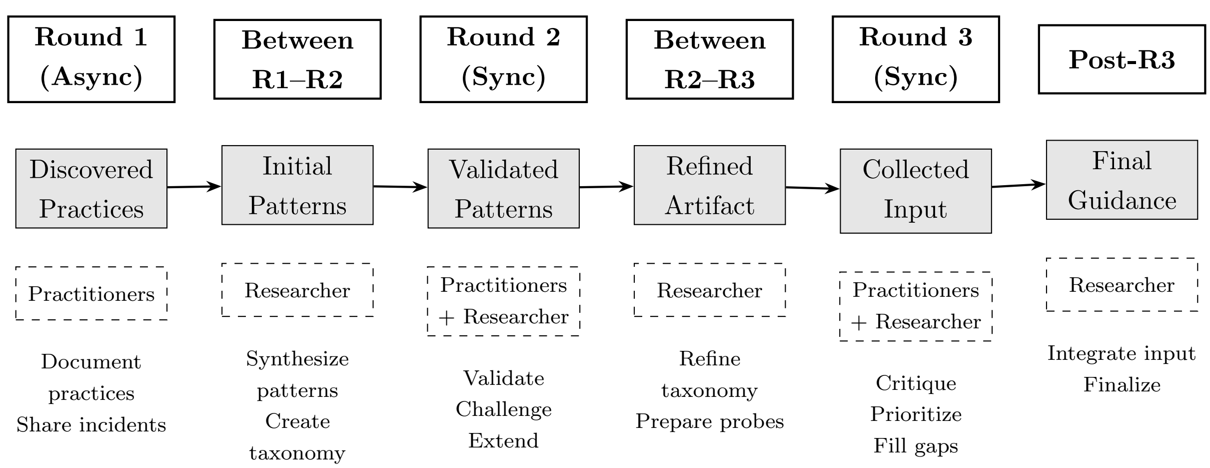

Figure 2. Emergent Artifact Development Process.

The guidance artifact evolves through six phases: practitioner discovery (Round 1), researcher synthesis (between rounds), collective validation (Round 2), researcher refinement (between rounds), practitioner critique (Round 3), and researcher finalization (post-Round 3). This approach aligns with grounded theory principles (Charmaz, 2014) where categories emerge from data rather than being predetermined.

4.2.3 Exploration and Development

The three rounds pursue sequential objectives. Round 1 focused on discovery: uncovering practices without researcher bias. Round 2 shifted toward convergence, validating patterns and identifying commonalities across practitioner experiences. Round 3 emphasized development, synthesizing collective expertise into practical guidance. This progression keeps prescriptive outputs grounded in descriptive findings.

4.3 Literature Review

This subsection describes how the theoretical foundation (Section 2) was built. The literature review followed a diverge-converge process (Botjes, 2020) that moved from broad exploration to focused selection, with the researcher's prior interests shaping the initial search direction.

The research began with a practical question: how does Generative AI (GenAI) impact team cognitive load for Enterprise Architecture (EA) practitioners? The researcher's background as a Team Topologies Advocate (Skelton & Pais, 2019) provided an initial lens. Skelton and Pais (2019) adapted Cognitive Load Theory (Sweller, 1988) to software delivery teams, arguing that team cognitive load determines delivery effectiveness. However, tracing this adaptation back to Sweller (1988)'s original Cognitive Load Theory revealed that the extrapolation from individual cognition to team-level load was more a conceptual stretch than a direct theoretical extension.

This gap led to Hutchins (1995)'s distributed cognition, which treats cognitive processes as distributed across people, artifacts, and environments rather than confined to individual minds. Distributed cognition proved a better theoretical fit, because it accounts for how cognitive work redistributes across human and artificial agents without requiring the individual-to-team extrapolation that Cognitive Load Theory demands. At the same time, EA practice proved too broad a scope. The rapid growth of GenAI coding tools was already reshaping software architecture and design (Peng et al., 2023), so the researcher narrowed the focus to one specific task: the Reference Architecture (RA) lifecycle, encompassing creating, updating, consuming, and learning from RAs. This combination of distributed cognition as theoretical lens and RA development as empirical scope shaped the theoretical foundation (Section 2).

4.3.1 Diverge-Converge Process

The literature search followed a diverge-converge process, adapted from Botjes (2020). As illustrated in Figure 3, the divergence phase creates choices by exploring broadly from primary sources outward to secondary and tertiary sources. The convergence phase makes choices by filtering sources through categories and keywords to determine relevance and scope.

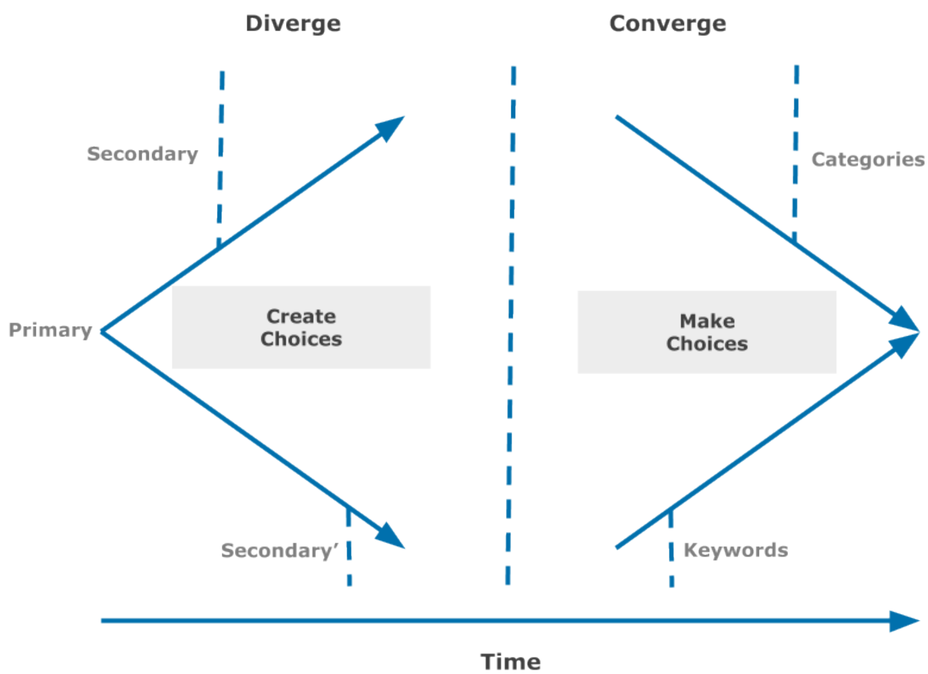

Figure 3. Diverge and Converge process. The divergence phase expands the search space from primary sources outward; the convergence phase filters by categories and keywords. Source: Botjes (2020).

Both phases were iterative. New sources discovered during convergence sometimes triggered additional divergence cycles as citation trails revealed unexpected connections. The path from Team Topologies (Skelton & Pais, 2019) to Cognitive Load Theory (Sweller, 1988) to distributed cognition (Hutchins, 1995) exemplifies this iterative process: each discovery opened new search directions before convergence narrowed the focus.

4.3.2 Search Strategies

Three complementary strategies populated the diverge-converge process.

Systematic database search. An EBSCO host search (June 2025) targeted top Information Systems journals using combinations of "enterprise architecture," "reference architecture," "generative AI," and "large language models"[2]. The near-zero results confirmed the research gap that motivates this study.

Backward and forward citation tracing. Foundational works served as primary sources for snowball searches. Starting from Zachman (1987) and Kotusev (2018) for Enterprise Architecture (EA), Hutchins (1995) and Hollan et al. (2000) for distributed cognition, Skelton and Pais (2019) and Sweller (1988) for Cognitive Load Theory, and Angelov et al. (2012) and Cloutier et al. (2010) for Reference Architectures, citation tracing identified additional sources across these domains. This strategy proved the most productive, yielding the majority of the theoretical foundation's sources.

Practitioner and grey literature. Industry reports from the International Energy Agency (International Energy Agency, 2025), McKinsey (McKinsey & Company, 2025), Microsoft (Microsoft, 2025), and Stack Overflow developer surveys (Stack Overflow, 2024, 2025) provided empirical context for GenAI adoption trends and infrastructure investment data not yet covered in peer-reviewed outlets.

Source selection prioritized peer-reviewed publications. Grey literature was used only for adoption statistics and infrastructure data. The resulting theoretical foundation (Section 2) synthesizes these sources into the five constructs and integrating lens that inform the conceptual model (Section 3).

4.4 Research Implementation

This subsection describes the expert panel selection, the hybrid round structure, data collection and analysis procedures, and detailed round protocols.

4.4.1 Expert Panel Selection

The expert panel was designed for 8–12 EA practitioners selected through purposive sampling (Patton, 2015). Eight participated in Round 1, ten of twelve invited joined Round 2, and seven of ten invited attended Round 3. Inclusion required current hands-on experience using GenAI in EA practice for at least six months and minimum five years as an EA practitioner to ensure foundational expertise. Candidates also needed active involvement in Reference Architecture development and willingness to participate across rounds. Consistent

[2]EBSCO host advanced search conducted on June 8, 2025, resulted in 0 publications using the query: ((enterprise architecture OR ea practice) AND (gen ai OR generative artificial intelligence OR llm OR large language models)) AND SO (MIS Quarterly Executive OR Information Systems Research OR Journal of the Association for Information Systems OR European Journal of Information Systems). Filters applied: peer-reviewed articles, publication date from January 1, 2020, to June 8, 2025.

with Kotusev (2018)'s finding that architectural artifacts involve distributed authorship, the panel deliberately includes professionals beyond the "enterprise architect" title. Participants hold roles including enterprise architect, solution architect, platform engineer, enterprise data architect, software architect, and freelance architectural consultant. This role diversity reflects how RA development unfolds in practice across organizational boundaries.

For Round 2, the panel was deliberately expanded from the original eight participants to twelve invitees (five returning from Round 1, seven new). This expansion served a methodological purpose: pattern validation benefits from perspectives not anchored to the respondents' own Round 1 submissions, reducing confirmation bias in category assessment. All new participants met the same inclusion criteria. Ten of twelve attended. For Round 3, all seven participants were returning from earlier rounds, ensuring continuity during the final convergence phase.

The panel composition seeks diversity across four dimensions: industry sectors (financial services, retail, technology, government), organizational size (SME to Fortune 500), geographic distribution within European timezone constraints, and varying levels of GenAI integration depth. This sample size aligns with Skulmoski et al. (2007), who suggest that 8–12 experts provide sufficient diversity while enabling meaningful synchronous interaction. This sampling strategy prioritises information richness and expert insight over statistical representativeness (Patton, 2015); the resulting panel captures frontier practice among active GenAI adopters rather than the breadth of the EA practitioner population.

4.4.2 Hybrid Delphi Round Structure

The study combines asynchronous and synchronous elements, moving from individual discovery to collective sensemaking. Figure 4 shows the complete process flow across all three rounds.

Round 1 (Asynchronous, January 2026): Individual exploration over two weeks

1. Participants received a discovery workbook via email
2. Completed structured sections at their own pace
3. Documented GenAI practices with rich contextual detail
4. Provided narrative accounts of critical incidents
5. Rated confidence and maturity of their practices

Rounds 2–3 (Synchronous, February–March 2026): Collective refinement

1. 120-minute online sessions using Zoom[3] for video conferencing and MeetingWizard[4]

[3]Zoom Video Communications: https://zoom.us/

[4]MeetingWizard is a cloud-based Group Support System (GSS) platform that structures sessions through agenda creation, anonymous brainstorming, structured voting, and real-time result aggregation, automati-

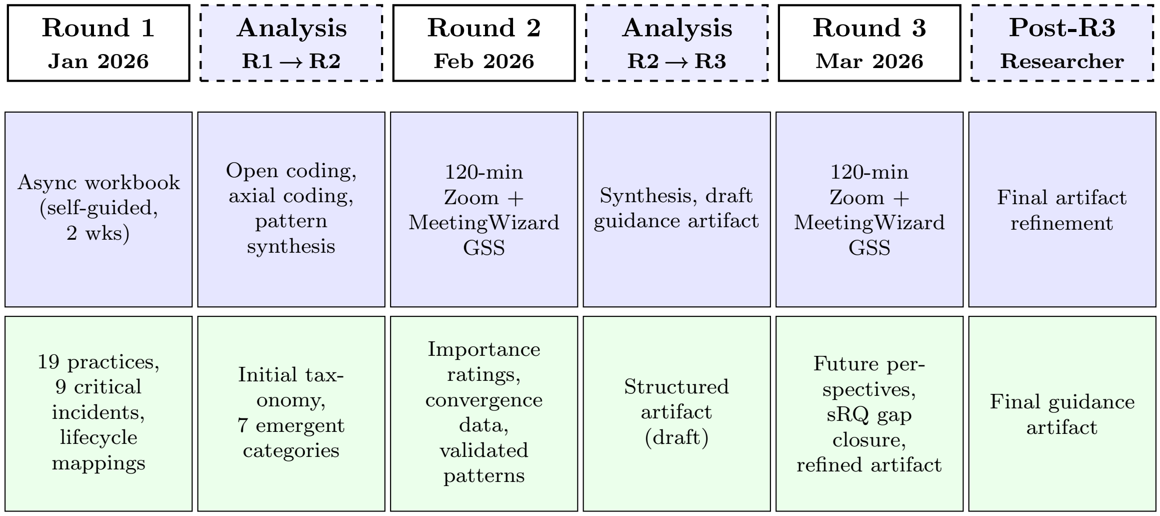

Figure 4. Delphi process flow with data progression. Three expert rounds (solid borders) alternate with researcher analysis phases (dashed borders). The middle row (blue) shows the method used in each phase; the bottom row (green) shows the key outputs.

for group deliberation

2. Real-time validation and exploration of emergent patterns
3. Dynamic consensus building with technological support
4. Progressive refinement of guidance artifact

4.4.3 Data Collection and Analysis

Each round yields progressively more structured data.

In the asynchronous Round 1, participants describe two to five discovered practices with step-by-step narratives and genesis stories. They map each practice to RA lifecycle phases and share three to five critical incidents (success and failure stories with supporting evidence). They also complete impact assessments covering time, quality, and role changes; document challenges and workarounds; and contribute emergent wisdom, including counter-intuitive discoveries and predictions about future practice evolution.

The synchronous Rounds 2–3 generated convergence data through structured GSS activities. Participants rated the importance of emergent practice categories on five-point scales and contributed to brainstorms on lifecycle mapping, practice changes, and trust calibration. Facilitated flipboard discussions explored boundary conditions and cross-role dynamics. Priority voting established collective importance weighting, while open-ended brainstorm responses captured why specific practices work or fail in particular contexts. Throughout these rounds, participants contributed refinements to the emerging guidance

cally generating reports that capture all discussions and outcomes. See https://www.meetingwizard.nl/en/

artifact.

4.4.4 Round Protocols

This subsection details the protocol for each round: Round 1 (asynchronous discovery), Round 2 (synchronous validation), and Round 3 (synchronous final refinement).

Round 1: Asynchronous Exploration (January 2026)

Objective: Open discovery of emerging practices without researcher framing

Two-Week Structure Round 1 gives participants two weeks to complete a discovery workbook at their own pace. No preliminary artifact or taxonomy is presented; the workbook uses sensitizing prompts that guide exploration without suggesting predetermined categories. Guided prompts ensure comparable data across participants, while free-text responses let practitioners describe practices in their own terms. Table 4 shows the timeline.

Table 4. Round 1 Timeline and Activities

Day	Activity	Deliverable	Expected Output
Day 1	Launch exploration	• Problem context document • Discovery workbook • Standard RA lifecycle	Participant orientation
Days 2–10	Self-guided discovery	Participants complete workbook sections	2–5 core practices per participant
Day 11–12	Reflection prompts	Additional prompts for unexplored areas	Completeness check
Day 13–14	Submission deadline	Completed workbooks	˜19 total practices
Days 15–25	Intensive analysis	Pattern emergence and synthesis	Initial taxonomy and guidance

Discovery Workbook Design The discovery workbook serves as the primary data collection instrument, structured to guide systematic exploration while avoiding constraining participants' responses. The workbook employs a progressive structure that moves from context setting through practice discovery to reflective sensemaking.

We deliberately framed the workbook scope as "Software Architecture / Solution Design work" rather than "Enterprise Architecture." This choice reflects a terminological broadening, not a scope expansion. Practitioners who create, update, and govern Reference

Architectures often describe this work as "solution design" or "software architecture" rather than "RA development." A strict EA framing risked excluding professionals who materially shape architectural artifacts but do not identify as enterprise architects: solution architects, platform engineers, and senior developers whose daily work involves Reference Architectures (Kotusev, 2018, see Section 2).

This terminological broadening introduces a validity tension. Practices discovered under a "Software Architecture" framing may extend beyond RA-specific work into general architectural practice, blurring the boundary with the research question's "EA practitioners" scope. Two instrument design features mitigate this risk. First, Part C of the workbook requires participants to map each practice to eight RA lifecycle phases, structurally anchoring reported practices to RA development regardless of the terminology used to invite participation. Second, the R2 validation session evaluated practices specifically in the context of RA development: ten participants from diverse architectural roles all recognized their practice in the workbook prompts. By meeting practitioners in their own terminology while anchoring responses to the RA lifecycle, the instrument captured practices that a narrower EA-only framing might have excluded without sacrificing scope specificity.

The workbook comprises seven parts. Part A establishes the participant's GenAI tooling, experience level, and organizational context. Part B (Practice Archaeology) forms the core discovery section, prompting participants to recall genesis moments and systematically document each named practice with its problem, step-by-step process, effectiveness rate, and failure modes. Part C asks participants to map their practices to eight RA lifecycle phases and identify tasks that must remain human-performed. Part D solicits three to five critical incident narratives with structured fields for context, action, result, insight, and supporting evidence. Part E captures impact reflections across three dimensions: time and effort, quality and completeness, and role evolution. Part F documents challenges encountered along with workarounds and capability wishes. Part G elicits emergent wisdom, including advice for colleagues, counterintuitive discoveries, and twelve-month predictions. This structure balances guided prompts for cross-participant comparability with free-text responses that allow practitioners to describe practices in their own terms. The complete workbook instrument is provided in B.

Data Collection Platform The workbook was deployed as a structured Word document distributed via email, providing:

- Completion over multiple sessions at the participant's own pace
- Structured sections guiding progressive exploration
- Rich text formatting for detailed descriptions

- Clear section delineation to encourage completeness

Outputs Round 1 yielded:

- 19 distinct practice descriptions (~2–3 per participant × 8 participants)
- 9 critical incident narratives with evidence
- Practice-to-lifecycle mappings across all workbook sections
- 12 challenges with workarounds
- 8 sets of emergent wisdom and predictions

This dataset underwent intensive analysis (Days 15–25) to identify emergent patterns, create an initial practice taxonomy, and develop the first version of the guidance artifact for validation in Round 2.

Round 2: Synchronous Pattern Validation (4 February 2026)

Objective: Validate emergent patterns from Round 1 and explore variations

Round 2 was conducted as a 120-minute synchronous session via Zoom[5] with MeetingWizard[6] GSS technology (see also Nieuwenhuis, 2025, for a recent application in EA research). Ten of twelve invited participants joined. The session presented emergent practice categories synthesized from Round 1 analysis for collective validation. The planned 15-step protocol was partially executed during the synchronous session: Steps 1–7 (opening, practice presentation, brainstorm on reactions and gaps, flipboard discussion on practice categories, importance voting, lifecycle mapping brainstorm, and before/after brainstorm) were completed in real time. Two remaining voting steps (Change Significance and Heuristic Agreement) were collected asynchronously afterward; all ten participants completed these votes. The session generated convergence data on practitioner agreement levels and cross-cutting patterns, providing input for the evolving guidance artifact.

Round 3: Synchronous Final Refinement (4 March 2026)

Objective: Gather future-oriented data, address remaining sub-research-question gaps, and collect input for the final guidance artifact

Round 3 was the final synchronous session, building on the validated practice categories from Round 2. Seven of ten invited participants attended the 120-minute session; all seven were returning participants from earlier rounds. The session was designed around three objectives. First, it aimed to elicit forward-looking perspectives on how GenAI practices in RA development may evolve. Second, it collected data for sub-research-questions not yet fully addressed, particularly sRQ3 on sensemaking and sRQ4 on systematization. Third, it gathered practitioner input to refine the draft guidance artifact. Procedural deviations from

[5] https://zoom.us/
[6] https://www.meetingwizard.nl/en/

the planned 16-step protocol are documented in Section 5.6.

Preliminary analysis of Rounds 1 and 2 had surfaced cross-role dynamics as an emergent theme. Kotusev (2018)'s finding that architectural artifacts involve distributed authorship provided theoretical grounding for this observation. This motivated three targeted R3 probes: how developer-level GenAI use informs architectural artifacts bottom-up, what traceability mechanisms practitioners employ when GenAI distributes authorship across roles, and where governance friction arises when non-architects produce architectural outputs.

4.5 Data Analysis Framework

The hybrid design requires distinct analytical approaches for the discovery phase versus the convergence phases:

4.5.1 Round 1 Discovery Analysis (Days 15–25)

The discovery analysis proceeds through four phases over Days 15–25. Phase 1 (data immersion) involves reading all workbooks, memoing first impressions, and creating participant profiles. Phase 2 (open coding) fractures data into discrete concepts using line-by-line, in-vivo, incident, process, and emotion coding (Charmaz, 2014). Phase 3 (pattern emergence) applies axial coding: identifying relationships between practices, comparing across participants, and clustering similar practices into preliminary categories.

These coding phases produced four coding dimensions that structure the Round 1 findings reported in Section 5: *interaction patterns* (how practitioners engage GenAI, from research assistant to deep integration), *distribution of cognitive tasks* (which tasks practitioners delegate versus retain), *engagement quality* (the degree of active oversight practitioners maintain over GenAI outputs), and *practice reconfiguration* (how established workflows changed with GenAI adoption). These dimensions emerged abductively: initial codes were derived inductively from practitioner language, then organized through constant comparison against the distributed cognition lens established in Section 3 (Charmaz, 2014; Patton, 2015). Together, the four dimensions provided a consistent analytical vocabulary for coding all 19 discovered practices and the cross-cutting patterns across the panel.

Phase 4 (synthesis) creates a practice taxonomy, maps practices to RA lifecycle phases, documents variation patterns, and prepares validation materials for Round 2. Three analytical decisions guide the process: practitioner language is preserved in category names, minority practices are kept even when mentioned by few participants, and contradictions are documented as variation points rather than treated as errors.

In addition to these coding categories, cross-role dynamics serve as an emergent analytical dimension. Round 1 revealed that practitioners routinely described practices crossing traditional role boundaries; what the coded analysis labels Pattern 4. Round 2's panel diversity reinforced this observation: inside-out workflows and collective decision capture surfaced precisely because participants held varied architectural roles. This dimension complements the existing coding categories (in-vivo, incident, process, emotion) as a thematic lens for identifying how GenAI redistributes architectural work across organizational roles. Hollan et al. (2000)'s distributed cognition framework (where cognitive processes distribute across individuals, artifacts, and environments) grounds this analytical choice theoretically.

4.5.2 Between-Round Analysis (Synchronous Rounds)

Round 2 → 3 Analysis

- Synthesized Round 2 importance ratings and variability profiles per practice category
- Confirmed seven practice categories; no category revisions required
- Prepared structured artifact with six elements per practice (description, lifecycle phases, guidance level, experience indicators, verification patterns, cross-role considerations)
- Designed targeted Round 3 probes based on emergent gaps (cross-role dynamics, trust calibration)

Round 3 Final Analysis

- Calculated consensus and variability metrics for all Round 3 voting items (heuristics, cross-role patterns, guidance priorities)
- Organized verification heuristics into tiers based on consensus strength
- Analyzed cross-role pattern significance; noted high variability as context-dependent rather than universal
- Integrated guidance element priority ratings into final artifact structure
- Finalized guidance artifact incorporating all three rounds of data

4.6 Quality Criteria

This subsection addresses methodological rigor, trustworthiness, and the study's limitations and delimitations.

4.6.1 Rigor in Exploratory-Convergent Delphi

Discovery validity in Round 1 is established through open-ended exploration where no researcher-imposed categories constrain what participants can report. Rich description through detailed narratives enables thick interpretation of practitioner experiences. Evidence grounding ensures that actual prompts and outputs support practitioner claims. Negative case analysis explicitly solicits failures and challenges alongside successes.

Convergence validity in Rounds 2–3 relies on four mechanisms. Member checking ensures participants validate interpretations at each round. Consensus metrics provide transparent reporting of agreement levels. Variation documentation preserves divergence without forcing artificial agreement, and progressive refinement enables iterative improvement across multiple rounds.

4.6.2 Trustworthiness Criteria

Delphi studies face established methodological critiques, particularly concerning researcher influence during synthesis, pressure toward artificial consensus, and arbitrary definitions of expertise (Linstone & Turoff, 1975; Okoli & Pawlowski, 2004). Following Lincoln and Guba (1985), this research addresses these risks through four trustworthiness criteria.

Credibility addresses the risk of researcher bias in interpreting and synthesizing participant responses. This study establishes credibility through prolonged engagement across three rounds over four months, allowing the researcher to develop deep familiarity with practitioner perspectives. Triangulation across multiple data types (narratives, ratings, and group discussions) reduces dependence on any single source. Member checking at each round ensures participants validate researcher interpretations before subsequent rounds build upon them. The analysis explicitly seeks negative cases, including failures and divergent views, rather than selecting only confirmatory evidence. Regular peer debriefing with the research supervisor provides external challenge to emerging interpretations.

Transferability addresses the risk that arbitrary or unclear expert selection limits the applicability of findings to broader contexts. The research provides thick description of participant organizational contexts, enabling readers to assess relevance to their own situations. Boundary conditions for guidance applicability are specified explicitly, acknowledging where practices transfer and where they do not. Documentation of practice variations across contexts preserves important distinctions rather than forcing premature generalization. The clear articulation of participant selection criteria in Section 4 allows readers to evaluate the expertise base underlying the findings.

Dependability addresses the risk of subjective analytical decisions lacking transparency. A comprehensive audit trail documents all analytical decisions, from initial coding through pattern synthesis to final guidance formulation. Code-recode reliability checks at two-week intervals ensure consistency in the researcher's interpretive framework over time. External advisor review of the coding framework provides independent validation of analytical categories. Version control for the guidance artifact (tracking its evolution across rounds) makes visible how collective input shaped the final output.

Confirmability addresses the risk that controlled feedback creates pressure toward artificial consensus, suppressing minority views. Reflexive notes captured at key analytical decision points document the researcher's evolving assumptions and their potential influence on interpretation. Direct participant quotes support all substantive interpretations, grounding claims in participant voice rather than researcher inference. Divergent views are explicitly preserved in the final guidance rather than eliminated in pursuit of false unanimity. Raw data remains available for audit, enabling independent verification of the analytical process.

4.6.3 Limitations

The methodology faces both methodological and scope limitations. Methodologically, discovery is limited to current practitioners, missing the perspective of non-adopters who tried and rejected GenAI integration. Practices are self-reported without direct observation, introducing recall bias. Social desirability bias likely persists despite anonymity measures. The synthesis between rounds requires researcher interpretation, creating opportunities for unintentional influence.

In terms of scope, the focus on the RA lifecycle excludes other EA activities where GenAI integration manifests differently. European timezone constraints limit global representation of the expert panel. The research captures current practices rather than future possibilities, providing a snapshot of an evolving phenomenon. Individual practices receive emphasis over organizational adoption patterns.

4.6.4 Delimitations

This research establishes purposeful boundaries to maintain focus and feasibility. The technology scope limits GenAI to LLM-based systems, excluding other AI types such as computer vision or predictive analytics. The artifact focus uses Reference Architecture as the primary lens rather than examining all EA deliverables.

The practice emphasis prioritizes discovery of what practitioners are currently doing over prescription of what they should do. The outcome orientation favors practical guidance for the EA community over purely theoretical contribution.

4.7 Ethical Considerations

The research adheres to university ethical guidelines and professional standards:

1. **Informed Consent**: Participants receive comprehensive information about the exploratory nature, time commitment, and data usage. Consent explicitly covers the iterative nature of the Delphi process.
2. **Anonymity Protection**: Individual practices are not attributed to specific participants. The Group Support System ensures anonymous contribution during synchronous sessions. Only aggregated findings appear in publications. Where individual participant quotations are presented, participants are identified by codes (P1–P4) to preserve anonymity.
3. **Intellectual Property**: Clear agreement that shared practices become collective knowledge for the EA community. Organizations retain rights to their specific implementations while contributing patterns to the common body of knowledge.
4. **Right to Withdraw**: Participants can exit the study at any point without consequence. Partial participation is valued and included where appropriate.
5. **Reciprocity**: All participants receive the final guidance document and summary findings. Acknowledgment offered (with permission) in publications.

Full disclosure of Generative AI tool usage throughout this research is provided in A.

4.8 Research Quality Criteria

Beyond the qualitative rigor criteria addressed above (credibility, transferability, dependability, confirmability), two additional frameworks inform the quality standards of this research.

Recker (2021) identifies four principles for scientific research: replicability, independence, precision, and falsifiability. This study addresses each as follows. **Replicability**: the methodology chapter documents protocols, instruments (Appendix B), and analytical procedures in sufficient detail for independent replication. **Independence**: the Delphi design uses anonymous input and controlled feedback to minimize researcher influence on participant responses. **Precision**: coding schemes, rating scales, and consensus thresholds are defined explicitly. **Falsifiability**: the study's claims are bounded by the data; negative cases and divergent views are preserved rather than suppressed, and limitations are stated explicitly.

The FAIR principles (Wilkinson et al., 2016) (Findable, Accessible, Interoperable, Reusable) guide the research data management strategy. Research instruments and the machine-readable guidance artifact are designed for reuse. The YAML artifact in particular enables computational processing by other researchers. Upon completion and grading, the research repository, including the machine-readable artifact,[7] will be made publicly accessible

[7] https://github.com/thiagoavadore/thesis_2024-2026_antwerp_thesis_GenAIPracticeVacuum/blob/

to support transparency and reproducibility.

4.9 Timeline and Logistics

Table 5 presents the research timeline spanning September 2025 through May 2026.

Table 5. Research Timeline

Phase	**Activities**	**Timeline**
Preparation	Recruitment, platform setup, workbook testing	September 2025
Round 1	Async discovery weeks + intensive analysis	January 2026
Round 2	Pattern validation session + analysis	February 2026
Round 3	Final round + synthesis	March 2026
Dissemination	Guidance publication and community sharing	April–May 2026

main/guidance-artifact.yaml

5 Results and Analysis

Introduction

This section presents findings from the three-round Delphi study (Section 4), organized by sub-research question. Within each subsection, findings follow the round progression to show how practitioner consensus evolved. The following subsections detail the results:

5.1 Delphi Panel and Convergence Overview

This subsection describes the expert panel composition across the three Delphi rounds and summarizes each round's procedural outcomes.

Table 6 presents the panel participation summary. Eight Enterprise Architecture (EA) practitioners completed asynchronous discovery workbooks in Round 1 (January 2026). Ten of twelve invited participants joined the synchronous Round 2 session (February 2026). Seven participants returned for the final synchronous Round 3 session (March 2026). For Round 2, the panel was expanded to twelve invitees (five returning, seven new) to introduce perspectives not anchored to Round 1 submissions. For Round 3, all seven participants were returning from earlier rounds, ensuring continuity in the final convergence phase.

Table 6. Panel Participation Summary

	Round 1	**Round 2**	**Round 3**
Format	Async workbook	Sync 120 min	Sync 120 min
Date	January 2026	4 February 2026	4 March 2026
Invited	8	12	10
Participated	8	10	7
Platform	Word workbook	Zoom[8] + MeetingWizard[9]	Zoom + MeetingWizard
Voted	–	10	7

Participants held diverse architectural roles including enterprise architect, solution architect, platform engineer, enterprise data architect, software architect, and freelance architectural consultant. This role diversity reflects how Reference Architecture (RA) development unfolds in practice across organizational boundaries, consistent with the broad "Enterprise Architecture (EA) practitioners" framing established in Section 4.4.1.

Round 1 focused on open discovery. Eight participants documented 19 distinct Generative AI (GenAI) practices, nine critical incidents, and twelve challenges with workarounds. No researcher-imposed categories constrained what participants could report, consistent with the exploratory Delphi orientation described in Section 4.

Round 2 shifted to pattern validation. The session presented six practice categories synthesized from Round 1 analysis. Participants voted on category importance, discussed boundary conditions, and proposed refinements. A seventh category (Governance and Compliance Checking) emerged during live discussion and was validated in real time. The session produced seven validated practice categories with importance ratings and convergence data on cross-cutting patterns.

Round 3 targeted remaining evidence gaps. The synchronous session executed Steps 1–14 of the planned 16-step protocol. Steps 1–12 proceeded as designed. Steps 13–14 (artifact discussion and guidance element voting) were compressed due to time constraints. Step 15 (Future Perspectives brainstorm) was not executed. Step 16 (SP2 Maturity Progression vote) was collected asynchronously after the session, with all seven participants responding. *Data Limitations* (Section 5.6) addresses the implications of these procedural variations.

5.2 sRQ1: GenAI Practices in RA Development

This subsection addresses sRQ1: *"What GenAI practices are EA Practitioners currently using when developing RAs?"* We present practice discovery from Round 1, category validation from Round 2, and cross-cutting patterns that emerged across both rounds. These findings form a two-level taxonomy: seven practice categories that describe task domains, and five cross-cutting patterns that describe behavioral universals transcending any single category.

5.2.1 Practice Discovery (Round 1)

Round 1 yielded 19 distinct GenAI practices from eight participants, each documenting two to three practices with step-by-step descriptions, genesis stories, and effectiveness assessments. The Round 1 coded analysis applied four coding dimensions: interaction patterns, distribution of cognitive tasks, engagement quality, and practice reconfiguration. These dimensions emerged through the open and axial coding process described in Section 4, following established qualitative coding procedures (Charmaz, 2014; Patton, 2015).

The practices concentrated in three RA lifecycle phases. Phase 4 (Target State Design) attracted the most practices, with seven of 19 mapped primarily to Research & Exploration activities. Phase 6 (Documentation & Communication) accounted for six practices across First Draft & Documentation and Stakeholder Communication. Phase 2 (Current State Analysis) contributed two practices in System/Codebase Analysis. Phases 1, 3, 5, 7, and 8 received no primary practice mappings, though some practices spanned multiple phases.

During the Round 2 session, the researcher clarified the breadth of RA work to participants. The researcher framed this as: "it can be a pattern, it can be a module... and you are not necessarily just creating, sometimes you are absorbing, going through it, reading, getting context" (R2 Transcript, researcher facilitation). The aim was to capture practitioner activities beyond artifact origination.

Interaction patterns varied across the panel. Research Assistant mode dominated, with seven of 19 practices involving GenAI as an information-gathering tool. First Draft Generator mode appeared in six practices where GenAI produced initial content for substantial human editing. Two participants operated in Cyborg mode with deep tooling integration via MCP servers, hooks, and skills. Two practices reflected a Junior Colleague framing with conversational, supervisory interaction. No participant delegated architectural decisions entirely to GenAI.

A key finding from Round 1 concerned the distribution of cognitive tasks. All eight participants delegated information gathering to GenAI while retaining judgment under ambiguity. Seven of eight described a hybrid synthesis pattern where GenAI drafts content and the human reconciles, validates, and edits. Organizational context (team dynamics, political constraints, historical decisions) remained exclusively human-retained across the entire panel.

5.2.2 Practice Category Validation (Round 2)

Round 2 validated the practice categories through collective deliberation. The researcher presented six categories synthesized from Round 1 analysis. During the flipboard discussion, participants identified a gap: governance and compliance checking was not represented in the original six categories. The panel proposed and validated a seventh category in real time, demonstrating the emergent nature of the Delphi process.

Table 7 presents the importance ratings for all seven validated categories. System/-Codebase Analysis received the highest mean rating with the lowest variability, indicating strong consensus on its importance for RA work. By contrast, Stakeholder Communication and Governance/Compliance showed the highest variability, reflecting context-dependent value across organizational settings.

Table 7. Practice Category Importance Ratings (Round 2, $N = 10$)

Practice Category	M	SD	**Var.%**	**Abst.**
System / Codebase Analysis	4.00	0.82	38	0
Research & Exploration	3.60	1.17	55	0
Adversarial Review / Stress Testing	3.60	0.97	45	0
Governance / Compliance Checking	3.50	1.27	60	0
First Draft & Documentation	3.10	1.10	52	0
Stakeholder Communication	3.10	1.37	65	0
Deep Workflow Integration	2.89	1.17	54	1

The consensus–variability pattern across categories showed a clear structure. System/Codebase Analysis exhibited both the highest importance and lowest variability, so directive guidance is feasible for this practice. Categories with higher variability (Stakeholder Communication, Governance/Compliance) call for guidance that remains contextual and adaptive. Deep Workflow Integration received the lowest mean rating and one abstention, consistent with its dependence on individual maturity and organizational tooling policies.

5.2.3 Cross-Cutting Patterns

The sRQ1 findings comprise a two-level taxonomy. The seven practice categories presented above constitute the first level: they describe *what* practitioners do, organized by task domain. Each category is context-dependent; its importance varies with organizational setting, as the consensus–variability profiles in Table 7 confirm. The five cross-cutting patterns below constitute the second level: they describe *how* practitioners approach GenAI work, regardless of task domain. The Round 1 coded analysis identified these patterns through constant comparison across participants (Charmaz, 2014); Round 2 discussions reinforced them.

Universal Verification Mandate (8/8 participants). All eight Round 1 participants described explicit verification practices. No participant accepted GenAI outputs without review. Verification strategies ranged from multi-round review cycles to line-by-line comparison with source documents, from knowledge-based assessment to treating all outputs as hypotheses. As P4 described: “Everything is treated as hypothesis, never conclusion” (R1 Workbook). P2 reported verifying “60–70% of specific claims” against primary sources (R1 Workbook). This universal pattern mapped to what we term *engaged augmentation*: a coding dimension that emerged from the Round 1 analysis to distinguish active oversight from passive delegation. Engaged augmentation characterized the entire panel.

Compression Not Generation (5/8 participants). Five participants framed GenAI’s value as compressing existing information rather than creating new architectural knowledge. As P2 described: “The value is not in generation but in compression. It compresses reading time for surveys” (R1 Workbook). P3 reinforced this framing: “The thinking is still mine; the AI accelerates expression” (R1 Workbook). This pattern reframes the distribution of cognitive tasks (Hutchins, 1995); GenAI handles information gathering while humans retain sensemaking.

Frontier Learning Through Failure (6/8 participants). Six participants described learning GenAI’s limitations through errors. P1 caught a retention period change from seven to five years only through verification. P2 identified a hallucinated benchmark that nearly entered a planning document. P3 encountered bounded context suggestions that were “technically logical but organisationally nonsensical” (R1 Workbook). These incidents demonstrated how practitioners discovered the boundary between tasks inside and outside GenAI capability through direct experience.

Organizational Context as Hard Boundary (7/8 participants). Seven participants identified organizational and sociotechnical factors as outside GenAI capability. P3 observed that "organisational context is nearly impossible to convey: team dynamics, political constraints, historical baggage" (R1 Workbook). P4 noted that GenAI "is not valuable for the sociotechnical aspects that determine whether an architecture will actually succeed" (R1 Workbook). This pattern identified a category of knowledge that practitioners consistently retained rather than delegated.

Role Identity Preservation (8/8 participants). All eight participants viewed GenAI as an efficiency tool, not a replacement for professional judgment. P4 stated: "I did this work for years without GenAI. Efficiency would decrease but effectiveness would not" (R1 Workbook). P3 described that he "would be slower but functional. The thinking is still mine" (R1 Workbook). This pattern indicated that professional identity remained intact despite significant workflow changes. These cross-cutting patterns, particularly the verification mandate and role identity preservation, set the stage for understanding how EA practice itself changed with GenAI adoption.

5.3 sRQ2: Changes to EA Practice

sRQ2 asks: *"What changes to EA practice are EA practitioners experiencing when RAs are developed using GenAI?"* The findings below cover practice changes from Rounds 1–2, cross-role dynamics that emerged across all three rounds, and a convergence assessment.

5.3.1 Practice Changes (Rounds 1–2)

Round 1 identified four dimensions of practice change:

1. **Creator to curator and judge.** Six of eight participants reported that their primary contribution moved from producing content to evaluating, editing, and validating AI-generated content.
2. **New skills emerged.** Prompt engineering, output evaluation, and trust calibration appeared across the panel as competencies that did not exist before GenAI adoption.
3. **Certain skills used less frequently.** Cold-start writing, exhaustive manual search, and cover-to-cover documentation reading declined.
4. **Artifact formats adapted.** Several practitioners shifted toward machine-readable formats such as Markdown and Mermaid diagrams.

Round 2 quantified these changes through significance voting. Table 8 presents the results. Verification burden shift received the highest rating with the lowest variability, indicating strong consensus that practitioners spend less time creating and more time validating. Compression not generation received the most contested rating, suggesting that not all practitioners frame GenAI's contribution as purely compressive.

Table 8. Change Significance Ratings (Round 2, $N = 10$)

Change Theme	M	SD	**Var.%**
Verification burden shift	4.40	0.70	33
Skills used less	3.90	0.57	26
Role evolution (creator → curator)	3.80	1.14	53
Compression not generation	2.90	0.99	47
New skills emerging	2.90	1.20	56

The low rating for "new skills emerging" ($M = 2.90$) alongside the high rating for "verification burden shift" ($M = 4.40$) revealed an interesting tension. Practitioners acknowledged that their work had changed substantially (verification) but were less inclined to characterize this change as requiring fundamentally new skills. This pattern indicated that practitioners treated verification as an extension of existing professional expertise, not a novel competency.

5.3.2 Cross-Role Dynamics (Rounds 1–2–3)

Cross-role dynamics emerged as an unanticipated theme across all three rounds. In Round 1, several practitioners described GenAI-enabled practices crossing traditional role boundaries: developers producing architectural artifacts bottom-up, solution architects generating decision records from meeting transcripts, and platform engineers contributing to enterprise-level governance. Round 2 reinforced these observations. The panel's role diversity surfaced inside-out workflows where developer-level GenAI use shapes EA artifacts before formal governance awareness.

Round 3 probed these dynamics through the "Ghost Architect" provocation: "If anyone with GenAI can produce architectural artifacts, who is the architect?" Seven participants contributed sixteen brainstorm responses across three probes (bottom-up workflows, traceability, governance friction). One participant observed: "The friction is real, and honestly, it's almost hard to admit. You spend 15 years building up knowledge... And then someone sits down with GenAI and produces something that looks pretty much the same in under an hour" (R3 Brainstorm). Another noted that "everybody can generate an architecture now, which is cool. But the one taking responsibility of the decision is in the end the architect" (R3 Brainstorm). These responses captured a shared tension: GenAI democratized artifact production, but accountability for architectural decisions remained with the architect.

Table 9 presents the cross-role pattern significance ratings from Round 3. Collective decision capture (meeting transcripts and requirements transformed into architecture decision records) received the highest mean rating ($M = 3.86$, $SD = 1.68$). However, all six patterns exhibited high variability (37–79%), indicating substantial disagreement about the significance of cross-role shifts. Governance patterns (inside-out and outside-in) received the lowest ratings, suggesting that GenAI had not yet materially affected governance structures in most participants' organizations.

Table 9. Cross-Role Pattern Significance (Round 3, $N = 5$–7)

Pattern	N	M	SD	**Var.%**
Collective decision capture	7	3.86	1.68	77
Top-down code generation	7	3.43	1.72	79
Role boundary blurring	7	2.86	1.57	72
Bottom-up artifact generation	7	2.57	1.27	58
Outside-in governance*	5	2.40	1.14	50
Inside-out governance	7	2.00	0.82	37

*Five of seven participants rated this item; two abstained, likely because this emergent pattern (proposed mid-session) was not applicable to their organizational context.

The high variability across all cross-role patterns reflected divergent organizational realities. Participants from organizations with mature governance structures reported minimal boundary shifts; those in more fluid environments described significant role redistribution. Cross-role dynamics appeared to depend on organizational factors rather than on GenAI adoption itself.

5.3.3 Convergence Assessment

Across the three rounds, several findings stabilized while others remained contested, consistent with the iterative convergence expected in Delphi designs (Von Der Gracht, 2012). The seven practice categories validated in Round 2 showed no challenges or revisions in Round 3, indicating stable consensus on the practice taxonomy. The universal verification mandate (8/8 in Round 1) was reinforced through Round 3's heuristic refinement, where cross-validation received the highest importance rating. The verification burden shift achieved the strongest consensus of any change theme ($M = 4.40$, variability 33%).

The SP2 maturity progression question, which could not be addressed during the synchronous session, was resolved through asynchronous data collection. The panel strongly agreed that practitioners naturally progress from basic to sophisticated GenAI integration ($M = 4.71$, variability 22%). At the same time, context received moderate agreement as a factor ($M = 3.86$, variability 49%), while the proposition that short experience can equal long experience was rejected ($M = 2.86$, variability 31%). The panel moderately agreed that maturity should be measured per practice category rather than per individual ($M = 3.86$, variability 49%). Table 10 presents the full results. Taken together, these ratings support a "both/and" interpretation: practitioners do progress over time, but progression is practice-specific and pace-modulated by organizational context.

Table 10. SP2 Maturity Progression Ratings (Async, $N = 7$)

Statement	M	SD	**Var.%**
Practitioners naturally progress from basic GenAI use to more sophisticated integration over time	4.71	0.45	22
Sophistication depends more on context than on individual experience	3.86	0.99	49
A practitioner with 6 months of experience can be as effective as one with 2 years, given the right context and guidance	2.86	0.64	31
Maturity should be measured per practice category, not per individual	3.86	0.99	49

Two areas remained empirically unresolved. First, the cross-role patterns showed variability too high to establish consensus, suggesting these dynamics are still emerging. Second, the relative importance of new skills versus existing expertise remained contested, with practitioners split on whether GenAI required fundamentally new competencies or merely extended existing professional judgment. The next subsection examines how practitioners made sense of these changes.

5.4 sRQ3: Sense-Making of Impacts and Trade-Offs

sRQ3 asks: *"How do EA practitioners make sense of impacts and trade-offs when adopting GenAI for RA development?"* Verification heuristics evolved across all three rounds, from personal strategies (Round 1) through candidate items (Round 2) to refined and rated heuristics (Round 3). Trust calibration stories and teachability findings follow.

5.4.1 Verification Heuristics (Rounds 1–2–3)

Verification heuristics evolved across all three rounds, progressively moving from personal strategies to collectively validated guidance. In Round 1, all eight participants described individual verification practices, but these varied widely in formality and specificity. The coded analysis identified five recurring heuristic patterns: never trust AI-generated numbers, organizational context is always a human job, treat outputs as hypotheses, verify line-by-line in compliance contexts, and AI usefulness decreases with domain specificity.

Round 2 formalized these patterns into five candidate heuristics and collected agreement ratings ($N = 10$). "Never trust AI-generated numbers" received the highest agreement ($M = 4.70$, $SD = 0.67$), followed by "In compliance contexts, verify every detail line-by-line" ($M = 4.20$, $SD = 0.63$). These two heuristics exhibited both high importance and low variability, indicating strong consensus.

Round 3 expanded and refined the heuristic set from five to eight items, incorporating participant-generated additions from the flipboard discussion. Table 11 presents the Round 3 importance ratings. Cross-validation of content (including source verification) received the highest rating with the lowest variability and no abstentions. Incremental trust building and context management also rated highly, both above 4.0. Output structure check received the lowest rating, with one abstention.

Table 11. Heuristic Importance Ratings (Round 3, $N = 5$–7)

Heuristic	N	M	SD	**Abst.**
Cross-validate content (source verification)	7	4.57	0.53	0
Incremental trust building	7	4.29	1.11	0
Context management	7	4.14	0.90	0
Domain expertise threshold	7	3.86	1.07	0
Learn through failure	5	3.40	0.89	2
Contradiction seeding	6	3.33	0.82	1
Simulate stakeholder reaction	5	3.00	1.58	2
Output structure check	6	2.83	1.33	1

A tier structure emerged from the ratings. The top tier ($M \geq 4.0$) comprised cross-validation, incremental trust building, and context management, all process-oriented heuristics that practitioners can systematically apply. The middle tier ($M = 3.0$–3.9) included domain expertise threshold, learn through failure, and contradiction seeding, heuristics that depend more on individual experience. The bottom tier ($M < 3.0$) contained output structure check, rated lowest and generating the most abstentions. This tier structure showed that process-oriented verification strategies achieved stronger consensus than experience-dependent ones.

5.4.2 Trust Calibration Stories (Round 3)

Round 3 collected 14 brainstorm responses from seven participants to the trust calibration prompt (participants could submit multiple responses): "Describe a specific moment when you had to calibrate trust in AI output." Two themes dominated the responses.

First, domain expertise as prerequisite appeared in multiple contributions. One participant described validating a solution design where GenAI "pointed to a folder that simply didn't exist. One key takeaway here: always verify the physical components where possible" (R3 Brainstorm). Another reported that "a lot of the hallucination issue can only be identified if you know things are wrong from experience already" (R3 Brainstorm). A third described an event-driven architecture where "the AI output was hallucinating capabilities of services that did not exist and wrongly explained retry logic for specific combinations" (R3 Brainstorm).

Second, the "junior colleague" framing recurred. One participant stated: "Think of AI as a junior architect and yourself as the senior architect" (R3 Brainstorm). Another described the trust-building process: "This is very much teachable: as normally we do this pre-AI when we train or have a buddy system. Learnt that using the right prompting and almost sharing the elements as if it were human betters the end output" (R3 Brainstorm). A third noted that dependency graph calculations "were actually better than our manual diagrams. We missed some links the LLM did not" (R3 Brainstorm). This indicated that trust calibration was bidirectional; practitioners sometimes discovered tasks where GenAI outperformed manual approaches, recalibrating trust upward rather than only guarding against errors.

5.4.3 Teachability of Calibration (Round 3)

The Round 3 flipboard discussion addressed a central tension: if verification depends on deep domain expertise and personal calibration, can it be taught? The panel's consensus was nuanced. Calibration is teachable, but it requires domain expertise as a foundation. One participant explained:

> Based on experience from the pre-AI era, you had to build a deep understanding before applying it. If the output of the AI is in line with the principles learned (scientifically grounded) and experience, at that point I trust it. It is not the LLM that is in charge. (R3 Brainstorm)

Several participants identified mechanisms for accelerating calibration in less experienced practitioners. Cross-checking with authoritative documentation, having beginners ask the AI to verify its own suggestions, and progressive exposure from low-stakes to high-stakes tasks all appeared as transferable strategies. One participant observed that "the person learning will have another learning trajectory compared with the persons from the pre-AI era. But in the end, the knowledge has to be built" (R3 Brainstorm).

One participant identified a gap in current reinforcement mechanisms. The observation pointed to the absence of structured feedback loops where practitioners learn which calibration decisions were correct after the fact. Without such reinforcement, calibration remains largely experiential, a limitation relevant to the guidance artifact design.

5.5 sRQ4: Systematization into Guidance

sRQ4 asks: *"How should effective GenAI practices for RA development be systematized into guidance for EA practitioners?"* Round 3 rated six candidate guidance elements, critiqued the draft artifact, and refined the artifact's structure.

5.5.1 Guidance Element Priorities (Round 3)

Round 3 presented the draft guidance artifact (v2.5) structured by practice type and asked participants to rate six candidate elements on importance. Table 12 presents the results. Practice description received the highest rating with the lowest variability, indicating near-unanimous agreement that describing what each practice is and when it applies is essential. Verification patterns (practice-specific trust calibration strategies) received the second-highest rating.

Table 12. Guidance Element Priority Ratings (Round 3, $N = 7$)

Guidance Element	M	SD	**Var.%**
Practice description	4.71	0.49	22
Verification patterns	4.14	0.69	31
Guidance level (directive vs. adaptive)	3.71	0.95	44
Experience indicators	3.43	0.98	45
Cross-role considerations	3.43	1.27	58
Lifecycle phase mapping	2.43	0.53	24

Lifecycle phase mapping received the lowest priority with low variability, indicating consensus that mapping practices to specific lifecycle phases was the least essential element. This finding was noteworthy given that the conceptual model in Section 3 emphasizes the RA lifecycle as a structuring lens. The panel's response suggested that practitioners valued practice descriptions and verification strategies over lifecycle positioning when consuming guidance.

Cross-role considerations exhibited the highest variability (58%) among the six elements, consistent with the contested cross-role dynamics observed in Section 5.3.2. This variability reflected genuine disagreement about whether cross-role considerations belong in practice-level guidance or at a higher organizational level.

Two additional elements surfaced during the verbal discussion but were not voted on: criticality-based applicability (applying practices differently based on system criticality) and proven prompt examples (high-quality prompt templates per practice). Both were noted for inclusion at the guidance level rather than per practice.

The SP2 maturity progression results (Table 10) further informed the guidance design. The panel's strong agreement on natural progression ($M = 4.71$) combined with moderate support for per-category measurement ($M = 3.86$) suggested that experience indicators should be differentiated per practice category rather than applied uniformly across the artifact.

5.5.2 Artifact Critique (Round 3)

Round 3 collected ten brainstorm responses from seven participants (participants could submit multiple responses) critiquing the draft artifact across four dimensions: structure, format, missing elements, and the directive-versus-adaptive distinction.

On structure, participants validated organizing by practice type rather than by lifecycle phase or experience level. One participant stated: "By practice seems fine, because if you do by lifecycle, then you will repeat yourself" (R3 Brainstorm). Another affirmed: "I think this is the right approach. The decision on how to apply AI will be done by use case as structured here" (R3 Brainstorm).

On format, participants expressed interest in multiple consumption modes. Several mentioned using it as a reference document paired with a checklist. One participant noted that "since the guide is already in md format it can be easily fed to an agent/assistant, so in addition to just a document there can be an assistant whom you could ask questions about this guide" (R3 Brainstorm). This response pointed toward agent-consumable guidance as a format consideration.

On missing elements, participants identified proven prompt examples and additional context for first-time readers as gaps. One participant noted: "It would need to be more verbose. At the moment it assumes a lot of background knowledge" (R3 Brainstorm). Another suggested including "proven, high-quality prompt samples" (R3 Brainstorm) per practice for less experienced practitioners.

On the directive-versus-adaptive distinction, one participant reported that "it took me a few minutes to figure out whether directive and adaptive were in the text" (R3 Brainstorm), suggesting the labeling needed clearer presentation.

5.5.3 Artifact Evolution Across Rounds

The guidance artifact progressed through four stages across the study, following the emergent artifact approach described in Section 4.

Initial taxonomy (between Rounds 1–2) emerged from the researcher's coded analysis of Round 1 data. It organized 19 discovered practices into an initial taxonomy mapped to RA lifecycle phases.

Validated patterns (between Rounds 2–3) incorporated Round 2 validation results. The seven practice categories replaced the initial taxonomy. Importance ratings and variability profiles informed a preliminary directive-versus-adaptive classification per category.

Structured artifact (presented in Round 3) structured each practice with six elements: description, lifecycle phases, guidance level, experience indicators, verification patterns, and cross-role considerations. This represented the researcher's synthesis of Rounds 1–2 findings into a concrete, reviewable artifact.

Final guidance (Section 6) incorporated Round 3 priorities, critique feedback, and the heuristic refinement data into the final guidance artifact. The panel's priority ratings (Table 12) directly informed which elements received emphasis.

Four data collection limitations warrant disclosure before the section summary.

5.6 Study Limitations Specific to Data Collection

Four data collection limitations affected the completeness of the findings presented in this section.

First, Round 3 Step 15 (Future Perspectives brainstorm) was not executed due to time constraints, and no asynchronous follow-up was conducted. This step was planned to collect forward-looking data on how GenAI practices in RA development may evolve. Its absence means the temporal dimension of the conceptual framework (Section 3.2.1) remains empirically underexplored.

Second, Step 16 (SP2 Maturity Progression vote) was collected asynchronously after the session rather than during it. All seven Round 3 participants responded, and the voting data is complete (Table 10). Three methodological caveats apply: (a) the vote was collected without the preceding group discussion (Step 15) that the protocol designed as reflective priming, (b) asynchronous collection eliminates the real-time deliberation and peer influence that characterize synchronous Delphi votes, and (c) participants voted individually rather than after collectively exploring future perspectives.

Third, Round 3 Steps 13–14 (artifact discussion and guidance element priority voting) were compressed due to time pressure. While all seven participants voted, the discussion preceding the vote was shorter than planned. This may have affected the depth of deliberation informing the priority ratings, though the voting data itself remains complete.

Fourth, participant counts varied across rounds and across voting items within rounds. Round 1 had eight participants, Round 2 had ten, and Round 3 had seven. Within Round 3, abstentions ranged from zero to two per item, particularly for the heuristic refinement vote. These variations limit the interpretive weight of individual items, though all items allow meaningful comparison as discovery indicators.

6 Guidance Artifact

Introduction

This section presents the guidance artifact answering sRQ4: *"How should effective GenAI practices for RA development be systematized into guidance for EA Practitioners?"* It turns the Delphi findings from *Results and Analysis* (Section 5) into prescriptive guidance that Enterprise Architecture (EA) practitioners can apply when using Generative AI (GenAI) in Reference Architecture (RA) development. The consensus-variability profiles from *Results and Analysis* (Section 5) directly inform where the artifact can offer directive recommendations and where guidance must remain adaptive. The artifact is organized by practice category (per the panel's preference), followed by cross-cutting guidance, governance adaptation, and a maturity model. The following subsections detail the artifact:

6.1 Introduction and Scope

The guidance artifact presented here is the final version, incorporating Round 3 priority ratings (Table 12), artifact critique feedback (Section 5.5.2), and heuristic refinement data (Table 11). The intended audience is Enterprise Architecture (EA) practitioners broadly defined: enterprise architects, solution architects, platform engineers, data architects, and senior engineers who participate in RA development. This scope matches the panel diversity described in Section 5.1.

The artifact is organized by practice category rather than by RA lifecycle phase or experience level, consistent with the panel's explicit preference (Section 5.5.2). One participant stated: "By practice seems fine, because if you do by lifecycle, then you will repeat yourself" (R3 Brainstorm). Lifecycle phase mapping received the lowest priority rating among all guidance elements ($M = 2.43$, Table 12), consistent with this choice.

Each practice category carries a guidance level (*directive*, *contextual*, or *adaptive*) based on panel agreement and variability patterns from Table 7. Categories where the panel agreed closely ($SD < 1.0$) receive directive guidance; those with moderate variability ($1.0 \leq SD \leq 1.2$) receive contextual guidance; and those with wide disagreement ($SD > 1.2$) receive adaptive guidance. Two overrides apply: Adversarial Review ($SD = 0.97$) is contextual rather than directive because its value depends on domain expertise, and Deep Workflow Integration ($SD = 1.17$) is adaptive because its low mean and abstention pattern point to adoption barriers that vary across settings. Practice descriptions ($M = 4.71$) and verification patterns ($M = 4.14$), the two highest-priority guidance elements, structure each practice block.

Each element carries a provenance label. *Delphi consensus* marks elements the panel voted on or explicitly validated. *Researcher synthesis* marks researcher interpretation anchored in Delphi data. *Researcher proposal* marks informed suggestions not validated by the panel. Where an element draws on both sources, both are indicated. External literature corroborating each practice is examined separately in Section 8.4.

6.2 Practice Catalog

Table 13 summarizes the seven validated practice categories with their consensus profiles, guidance levels, and primary verification patterns. The following subsections detail each practice.

Throughout the catalog, practice descriptions and verification patterns carry Delphi consensus; experience indicators are researcher proposals.

System/Codebase Analysis *[Directive].* This practice involves using GenAI to

Table 13. Practice Catalog Summary

Practice Category	*M*	*SD*	**Guidance Level**	**Primary Verification Pattern**
System / Codebase Analysis	4.00	0.82	Directive	Cross-validate against source systems
Research & Exploration	3.60	1.17	Contextual	Verify claims against primary sources
Adversarial Review / Stress Testing	3.60	0.97	Contextual	Domain expertise threshold check
Governance / Compliance	3.50	1.27	Adaptive	Line-by-line verification in compliance contexts
First Draft & Documentation	3.10	1.10	Contextual	Cross-check references and facts
Stakeholder Communication	3.10	1.37	Adaptive	Validate organizational context manually
Deep Workflow Integration	2.89	1.17	Adaptive	Incremental trust building

analyze existing systems, codebases, and architectural artifacts to build understanding of the current state. Practitioners should apply this practice during Phase 2 (Current State Analysis) of the RA lifecycle, particularly for legacy system comprehension, dependency mapping, and cross-repository analysis. System/Codebase Analysis received the highest importance rating with the lowest variability (Table 7), indicating strong consensus that this is the most valuable GenAI application in RA work.

Verification pattern: Cross-validate GenAI-generated system descriptions against source systems directly. As the Delphi panel established, all participants delegated information gathering to GenAI while retaining judgment (Section 5.2.1). For codebase analysis specifically, practitioners should verify that referenced components, services, and dependencies exist in the actual system. One Round 3 participant reported that GenAI "pointed to a folder that simply didn't exist" (Section 5.4.2), demonstrating why physical constraint checking is essential.

Experience indicators: Novice practitioners should start with well-documented codebases where verification is straightforward. Advanced practitioners extend this practice to legacy systems and cross-repository analysis where GenAI's compression value is highest.

Research & Exploration *[Contextual].* This practice involves using GenAI as a research assistant for exploring design alternatives, technology options, and architectural trade-offs during target state design. Seven of 19 Round 1 practices mapped to Phase 4

(Target State Design) in research-assistant mode (Section 5.2.1).

Verification pattern: Verify claims, statistics, and technology recommendations against primary sources. The cross-validation heuristic received the highest importance rating across all heuristics ($M = 4.57$, Table 11). P2 reported verifying "60–70% of specific claims" against primary sources (R1 Workbook). Practitioners should treat GenAI research outputs as hypotheses requiring confirmation, not as authoritative findings.

Experience indicators: Start with well-scoped questions verifiable against established sources; progress to multi-model workflows and iterative refinement.

Adversarial Review / Stress Testing *[Contextual].* This practice involves deliberately using GenAI to challenge, critique, and stress-test architectural decisions. The Delphi panel validated contradiction seeding as a verification heuristic ($M = 3.33$, Table 11), and domain expertise threshold rated $M = 3.86$. Adversarial Review requires the practitioner to actively elicit critique rather than accept GenAI's default collaborative stance.

Adversarial Review encompasses a family of techniques that practitioner discourse calls *Architecture Stress Testing*. Table 14 presents seven named sub-practices identified through researcher synthesis. The Delphi panel validated contradiction seeding and domain expertise threshold as techniques; the remaining techniques were identified from practitioner discourse and are corroborated in Section 8.4.

Table 14. Architecture Stress Testing Techniques

Technique	Description
AI Sparring Partner	Iterative adversarial dialogue where GenAI challenges design rationale
Devil's Advocate Prompting	Instructing GenAI to argue against the proposed architecture
Decision Stress Testing	Sequential methodology: identify blind spots, steelman alternatives, conduct pre-mortem
AI Pre-Mortem	"The architecture already failed; why?" Prospective hindsight analysis
Skeptical Reviewer	Role-based persona critique simulating a senior architect review
Architectural Gap Analysis	Structural completeness analysis against standards and patterns
Epistemic Stress Testing	Challenges confidence levels behind decisions, not the decisions themselves

Verification pattern: Adversarial Review requires domain expertise as a prerequisite. The panel confirmed that "a lot of the hallucination issue can only be identified if you know things are wrong from experience already" (Section 5.4.2). Practitioners should evaluate

GenAI-generated critiques against their own domain knowledge, recognizing that GenAI lacks the ability to model strategic adversarial behaviour across organizational boundaries.

Experience indicators: Novice practitioners should start with structured techniques such as AI Pre-Mortem and Skeptical Reviewer, which provide clear prompting templates. Advanced practitioners develop intuition for when GenAI critique reflects genuine design weaknesses versus pattern-matching artifacts.

Governance/Compliance Checking *[Adaptive].* This practice involves using GenAI to verify architectural artifacts against governance standards, compliance requirements, and organizational policies. Governance/Compliance emerged during live Round 2 discussion as the seventh category (Section 5.2.2), with high variability ($SD = 1.27$) reflecting context-dependent value across organizational settings. The adaptive guidance level acknowledges that governance structures vary substantially across organizations.

Verification pattern: In compliance contexts, verify every detail line-by-line. The Delphi panel rated this heuristic $M = 4.20$ with the lowest variability among Round 2 heuristics. GenAI can scan documents against regulatory requirements, but practitioners must verify completeness and accuracy manually. Compliance errors carry disproportionate organizational risk.

Experience indicators: Begin with low-risk compliance contexts; progress to automated governance pipelines where organizational maturity permits.

First Draft & Documentation *[Contextual].* This practice involves using GenAI to produce initial drafts of architectural documentation (solution designs, architecture decision records (ADRs), technical specifications, and assessment reports) for subsequent human editing and validation. Six Round 1 practices employed the First Draft Generator interaction mode (Section 5.2.1). The Compression Not Generation framing (Section 5.2.3) applies directly: GenAI compresses existing information into structured documents rather than creating novel architectural knowledge.

Verification pattern: Cross-check all references, links, and factual claims in generated documentation. The panel's experience included hallucinated benchmarks and non-existent references that nearly entered planning documents (Section 5.2.3). Practitioners should treat generated drafts as starting points requiring architectural judgment, not finished artifacts.

Experience indicators: Start with well-structured document types (meeting minutes, status reports) before progressing to architecture decision records and metaprompt libraries.

A related emerging practice, Spec-Driven Development, extends this documentation-first principle; its external evidence base is examined in Section 8.4.

Stakeholder Communication *[Adaptive].* This practice involves using GenAI to translate architectural content for different audiences (executive summaries, team-specific views, non-technical explanations) and to synthesize stakeholder inputs into structured architectural artifacts. Stakeholder Communication exhibited the highest variability of all categories ($SD = 1.37$, Table 7), indicating substantial disagreement about GenAI's value for this purpose.

Verification pattern: Validate organizational context manually before communicating GenAI-assisted outputs. The panel established that organizational context (team dynamics, political constraints, historical decisions) remains exclusively human-retained knowledge (Section 5.2.3). GenAI-generated stakeholder communications risk being "technically logical but organisationally nonsensical" (P3, R1 Workbook) if organizational context is not carefully injected.

Experience indicators: Limit initial use to summarization tasks (meeting minutes, document distillation) where source material constrains outputs; extend to audience-specific prompting and RAG-based approaches as familiarity grows.

Deep Workflow Integration *[Adaptive].* This practice involves embedding GenAI into the practitioner's daily workflow through IDE integration, agent context files, custom hooks, Model Context Protocol (MCP) servers, and multi-agent orchestration. Deep Workflow Integration received the lowest mean rating ($M = 2.89$, one abstention, Table 7), consistent with its dependence on individual maturity and organizational tooling policies.

The low rating likely reflects adoption barriers (cognitive load of tooling setup, organizational resistance, and infrastructure prerequisites) rather than low practice value. Two panel members operated in Cyborg mode with deep tooling integration (Section 5.2.1), showing that practitioners who overcome these barriers achieve sustained workflow benefits.

Verification pattern: Build trust incrementally. The incremental trust building heuristic rated $M = 4.29$ (Table 11), making it the second-highest heuristic and particularly relevant for deep integration where automated workflows reduce human oversight touchpoints. Practitioners should progressively expand GenAI autonomy from low-stakes tasks to higher-stakes ones as calibration develops.

Experience indicators: Novice practitioners should begin with single-tool integration (IDE assistant) before progressing to custom context files and agent orchestration. Advanced practitioners design multi-agent architectures with specialized agents for different architectural tasks.

6.3 Cross-Cutting Guidance

The practice catalog addresses individual practice categories. Three cross-cutting concerns apply across all seven categories: trust calibration, context engineering, and skills atrophy mitigation. These concerns emerged from the cross-cutting patterns identified in

Section 5.2.3 and the verification heuristics refined across all three Delphi rounds (Section 5.4.1).

6.3.1 Trust Calibration Framework

Trust calibration is the foundational cross-cutting concern. The Delphi panel established a universal verification mandate: all eight Round 1 participants described explicit verification practices, with no participant accepting GenAI outputs without review (Section 5.2.3). The cross-validation heuristic received the highest importance rating of any heuristic ($M = 4.57$, $SD = 0.53$, Table 11).

Table 15 presents a trust spectrum synthesized from Delphi data and external evidence (Researcher synthesis). The panel validated the underlying verification heuristics, but did not vote on the specific trust levels or their ordering. The spectrum structure is a researcher interpretation anchored by the panel's verification experiences and the organizational context boundary established in Section 5.2.3. This spectrum translates the verification heuristics from Section 5.4.1 into task-level trust guidance, connecting the empirical sense-making patterns (sRQ3) to the systematized guidance framework (sRQ4).

Table 15. Trust Spectrum for GenAI in RA Development

Trust Level	Task Type	Verification Approach
Highest	Information compression: summaries, boilerplate, dependency graphs	Spot-check representative samples
High	Documentation drafts, concept explanation, test generation	Review structure and key claims
Medium	Code refactoring, common patterns, meeting synthesis	Verify against organizational standards
Medium-Low	Design trade-off exploration, technology recommendations	Cross-validate with primary sources
Low	Architecture decisions, novel business logic	Independent domain expert review
Very Low	Security-critical code, compliance assessment	Line-by-line verification
Lowest	Organizational judgment: politics, team dynamics, legal	Retain entirely; do not delegate

Trust calibration is bidirectional. The Delphi panel identified instances where GenAI outperformed manual approaches; dependency graph calculations "were actually better than our manual diagrams" (R3 Brainstorm, Section 5.4.2). Practitioners should recalibrate trust upward when evidence warrants, not only guard against errors. The "junior colleague" mental model recurred across the panel: GenAI outputs merit the same supervisory attention as work from a capable but inexperienced team member.

Two prerequisites anchor the framework. First, trust calibration is teachable but requires domain expertise as a foundation (Section 5.4.3). Practitioners "had to build a deep understanding before applying it" (R3 Brainstorm). Second, calibration improves through the frontier learning pattern: six of eight participants discovered GenAI's limitations through direct experience with errors (Section 5.2.3). These findings align with Dell'Acqua et al. (2023), who demonstrated that knowledge workers operating on GenAI's "jagged technological frontier" must learn where AI capability boundaries fall through practice.

6.3.2 Context Engineering for EA

Rajasekaran et al. (2025) defines context engineering as "the set of strategies for curating and maintaining the optimal set of tokens during LLM inference." The Delphi panel validated the importance of this practice through the context management heuristic ($M = 4.14$, $SD = 0.90$, Table 11), the third-highest rated heuristic.

Context engineering narrows the distance between what practitioners know and what GenAI can access. Seven of eight participants identified organizational context (team dynamics, political constraints, historical decisions) as outside GenAI capability (Section 5.2.3). Effective context engineering does not eliminate this boundary but narrows it by making more organizational knowledge machine-accessible.

For EA practitioners, context engineering manifests in three areas (the panel validated context management as important; the three-part structure is researcher synthesis). First, *agent context files*: project-level configuration files such as CLAUDE.md and AGENTS.md that encode architectural standards, design principles, and organizational constraints (Chatlatanagulchai et al., 2025). An empirical study of 2,303 such files found that architecture content appeared in 67.7% of repositories, with files undergoing multiple commits and evolving every one to three days. These files function as a new category of architectural artifact: both human-readable documentation and machine-executable governance.

Second, *strategic information ordering* mitigates the "lost in the middle" effect, where language models attend less to information in the middle of long contexts (Liu et al., 2024). Practitioners should place critical architectural constraints at the beginning and end of context, with supporting detail in the middle. Modular rules files (separate context documents for different architectural concerns) keep individual contexts focused and within effective processing ranges.

Third, *context scope management* addresses what to include and what to exclude.

Effective context includes: architectural principles and constraints, relevant standards and patterns, system-specific technical context, and decision history (ADRs). Effective context excludes: organizational politics and interpersonal dynamics (which GenAI cannot reliably process), exhaustive historical documentation (which dilutes signal), and ambiguous or contradictory requirements (which amplify hallucination risk).

6.3.3 Anti-Atrophy Strategies

Practitioners risk losing skills they stop exercising. The skills GenAI automates (cold-start writing, exhaustive manual search, cover-to-cover documentation reading) are the same skills through which practitioners built their expertise (Kabashkin, 2025). The Delphi panel rated "skills used less" as the second-most significant practice change ($M = 3.90$, Table 8), confirming that practitioners recognize this dynamic in their own experience.

Three deliberate friction practices mitigate competency atrophy (the panel validated the atrophy concern; the three mitigation strategies are researcher proposals). First, *AI-free design sessions*: periodically conducting architectural work without GenAI assistance to maintain foundational skills. This practice is analogous to how pilots train for manual flight despite autopilot capabilities. Second, *attempt independently first*: engaging with a problem for 15–30 minutes before consulting GenAI, ensuring the practitioner develops and maintains their own mental model before seeing an AI-generated one. Third, *pair programming mindset*: treating GenAI interaction as collaborative work rather than delegation, maintaining active cognitive engagement throughout. The practitioner drives architectural thinking while GenAI handles information retrieval and expression.

These strategies connect to the role identity preservation pattern (Section 5.2.3). All eight Delphi participants viewed GenAI as an efficiency tool rather than a replacement for professional judgment. As P4 stated: "I did this work for years without GenAI. Efficiency would decrease but effectiveness would not" (R1 Workbook). Anti-atrophy practices operationalize this stance by ensuring that practitioners maintain the independent capability that underpins effective GenAI supervision.

6.4 Governance Adaptation

Governance adaptation for GenAI in RA development rests on thinner Delphi evidence than the practice catalog and cross-cutting guidance. Governance patterns received the lowest significance ratings among cross-role dynamics (Table 9), and Governance/Compliance Checking exhibited the second-highest variability ($SD = 1.27$, Table 7). We present three governance themes that emerged from Delphi data triangulated with external evidence, while acknowledging that most organizations have not yet formally adapted their governance structures.

Gate-to-guardrail transition *(Researcher synthesis).* Traditional EA governance operates through periodic architectural review boards (ARBs) that evaluate decisions at defined stage gates. GenAI's acceleration of artifact production (practitioners producing drafts in minutes rather than days) challenges this periodic model. The Delphi panel's low rating for inside-out governance ($M = 2.00$, Table 9) suggests that most organizations have not yet confronted this shift formally. Ettinger (2025) argues that EA frameworks have significant limitations addressing GenAI requirements, noting the tension between innovation enablement and compliance maintenance. In our view, organizations should consider transitioning from periodic gate reviews to continuous guardrail enforcement, where architectural constraints are encoded and checked automatically rather than evaluated episodically.

Architecture-as-Code enforcement *(Researcher proposal).* Encoding architectural constraints as machine-readable rules (TypeScript validators, fitness functions, CI/CD checks) enables continuous enforcement without manual review bottlenecks. This approach bridges the Governance/Compliance practice category with Deep Workflow Integration by embedding governance directly into development workflows. When architectural rules are machine-readable, GenAI agents can read and apply them before generating code, reducing violation rates. Early implementations demonstrate executable ADRs where architectural decisions become automated rules in build pipelines (Ford et al., 2023; Zdun et al., 2013). This represents a shift from documenting governance intent to operationalizing it.

AI disclosure and provenance *(Researcher synthesis).* As GenAI-generated artifacts become prevalent, governance frameworks need mechanisms for identifying which components were AI-generated or AI-augmented. The "Ghost Architect" dynamic (where "everybody can generate an architecture now," Section 5.3.2) creates traceability challenges. AI disclosure policies flag AI-generated components for calibrated review, enabling review boards to apply appropriate scrutiny. Provenance tracking becomes particularly important for compliance-sensitive domains where the origin of architectural decisions carries regulatory implications.

6.5 Maturity Model

The Delphi data revealed that practitioners operate at different levels of GenAI integration, from ad-hoc experimentation to deep workflow embedding. Round 1 workbooks showed this variation clearly: two participants operated in Cyborg mode with MCP servers and custom hooks, while others used chat-based interaction exclusively (Section 5.2.1). Table 16 presents a four-stage maturity model derived from this observed variation, mapped to the practice catalog and cross-cutting guidance. The four-stage structure is a researcher proposal; the progression assumption that practitioners naturally advance from basic to sophisticated integration has Delphi consensus ($M = 4.71$, Table 10). This structure follows established maturity model conventions. Becker et al. (2009) propose a procedure model for theoretically founded maturity model development, noting that over a hundred such models have been created across IT management domains. Pöppelbuß and Röglinger (2011) characterize maturity models as staged representations of anticipated evolutionary paths, where each stage builds on the previous one. A recognized critique of such models is that the assumed sequential progression lacks empirical causal validation (Pöppelbuß & Röglinger, 2011; van Steenbergen, 2011); Section 6.5 addresses this by grounding the four stages in observed practitioner variation rather than prescriptive theory.

Table 16. Maturity Model for GenAI in RA Development

Stage	Label	Characteristics	Observable Indicators
1	Experimental	Ad-hoc GenAI use for individual tasks; chat-based interaction; no structured verification	Occasional use for research or drafting; no shared practices; trust calibration through trial and error
2	Systematic Personal	Structured personal workflows; context engineering basics; deliberate verification	Personal prompt libraries; agent context files; anti-atrophy practices; consistent trust spectrum application
3	Team-Level Integration	Shared practices across team; team-level agent context files; Architecture-as-Code adoption	Shared context files in version control; team verification standards; collective decision capture workflows
4	Organizational Embedding	Governance adaptation; enterprise context engineering; agentic workflows	Automated architectural guardrails; AI disclosure policies; multi-agent orchestration; enterprise MCP infrastructure

The SP2 maturity progression vote, collected asynchronously after the Round 3 session (Table 10), provides partial empirical grounding. The panel rated natural progression from basic to sophisticated integration at $M = 4.71$ (variability 22%), supporting the model's progression assumption. The proposition that short experience can equal long experience scored $M = 2.86$, indicating that progression requires sustained engagement. Per-category measurement scored $M = 3.86$ (variability 49%), suggesting practitioners advance at different rates across practice categories. Context scored the same ($M = 3.86$) but appears to modulate pace rather than replace experience.

The model is diagnostic, not prescriptive. Practitioners can use it to locate their current practice and identify growth areas; progression through all four stages is not necessary for every context. The progression assumption has panel support, though the four-stage structure itself remains researcher-proposed. *Data Limitations* (Section 5.6) addresses the methodological caveats, and the *Discussion* (Section 8) interprets the implications for EA maturity theory.

7 Conclusion

Introduction

This section draws conclusions from the findings, identifies contributions to practice, and acknowledges limitations. Theoretical contributions and future research directions are presented in the *Discussion* (Section 8). The following subsections detail these conclusions:

7.1 Research Questions Revisited

The central research question asked: *What practices do Enterprise Architecture (EA) practitioners adopt when using Generative AI (GenAI) for Reference Architecture (RA) development, and how can these be systematized into guidance?* The Design-Oriented Delphi methodology, conducted across three rounds with eight to ten participants, produced convergent answers through four sub-questions.

sRQ1: What GenAI practices are EA practitioners currently using? The Delphi panel identified and validated seven practice categories: System/Codebase Analysis, Research & Exploration, Adversarial Review, Governance/Compliance Checking, First Draft & Documentation, Stakeholder Communication, and Deep Workflow Integration (Section 5.2). *Conclusion:* EA practitioners have developed structured, category-specific GenAI practices within months of tool availability. The strong consensus on System/Codebase Analysis and the weaker consensus on Stakeholder Communication suggest that GenAI adds more value where less organizational context is needed.

sRQ2: What changes to EA practice are practitioners experiencing? Three interrelated shifts emerged: the verification burden shift, the creator-to-curator transformation, and changing cross-role dynamics (Section 5.3). *Conclusion:* GenAI integration restructures the practitioner's role from content creator to content curator. The verification burden shift is the most significant change, indicating that quality assurance, not content generation, is the primary challenge.

sRQ3: How do practitioners make sense of impacts and trade-offs? The panel articulated eight verification heuristics in two tiers, anchored by trust calibration and the junior colleague mental model (Section 5.4). *Conclusion:* Practitioners navigate GenAI integration through calibrated trust rather than binary accept/reject decisions. The junior colleague metaphor provides a shared cognitive frame for positioning GenAI contributions within the team.

sRQ4: How should practices be systematized into guidance? The guidance artifact (Section 6) organizes recommendations at three levels: directive, contextual, and adaptive (Section 5.5). *Conclusion:* Effective guidance must respect the variability in practitioner consensus. A single set of prescriptive rules would not serve a phenomenon where GenAI value ranges from consistently high (codebase analysis) to heavily context-dependent (stakeholder communication).

Table 17 summarizes these conclusions. Figure 5 maps them back onto the conceptual framework introduced in Section 3.2, showing how each sub-question produced a distinct finding within the practice evolution cycle.

Table 17. Summary of Research Question Conclusions

Sub-RQ	Question Focus	Conclusion
sRQ1	Current GenAI practices	Seven validated practice categories spanning RA development, from System/Codebase Analysis (strongest consensus) to Deep Workflow Integration (emerging).
sRQ2	Changes to EA practice	Three practice evolution shifts: verification burden shift, creator-to-curator transformation, and changing cross-role dynamics.
sRQ3	Sense-making of impacts	Eight verification heuristics in two tiers (structural and domain), anchored by trust calibration and the junior colleague mental model.
sRQ4	Systematization into guidance	Three-tiered guidance artifact: directive (high consensus), contextual (moderate), and adaptive (high variability), organized by practice category.

Overall conclusion: EA practitioners are developing structured, category-specific practices for GenAI integration, and these practices can be systematized through consensus-validated guidance that respects the variability inherent in context-dependent architectural work.

7.2 Contributions

Theoretical contributions (extending distributed cognition, practice theory, and the junior colleague mental model as calibration heuristic) are developed in Section 8.6. Here we summarize the practical contributions.

7.2.1 Contributions to Practice

Five practical contributions emerge from these results. First, the practice taxonomy describes what practitioners actually do with GenAI, not what the technology can theoretically do. Each of the seven categories maps to specific RA development activities and gives organizations a shared vocabulary for discussing GenAI integration.

Second, the consensus–variability profiles reveal where GenAI adds consistent value and where its contribution remains contested. System/Codebase Analysis ($M = 4.00$, $SD = 0.82$) represents a zone of reliable augmentation, while Stakeholder Communication ($M = 3.00$, $SD = 1.37$) marks the organizational context boundary where GenAI value depends

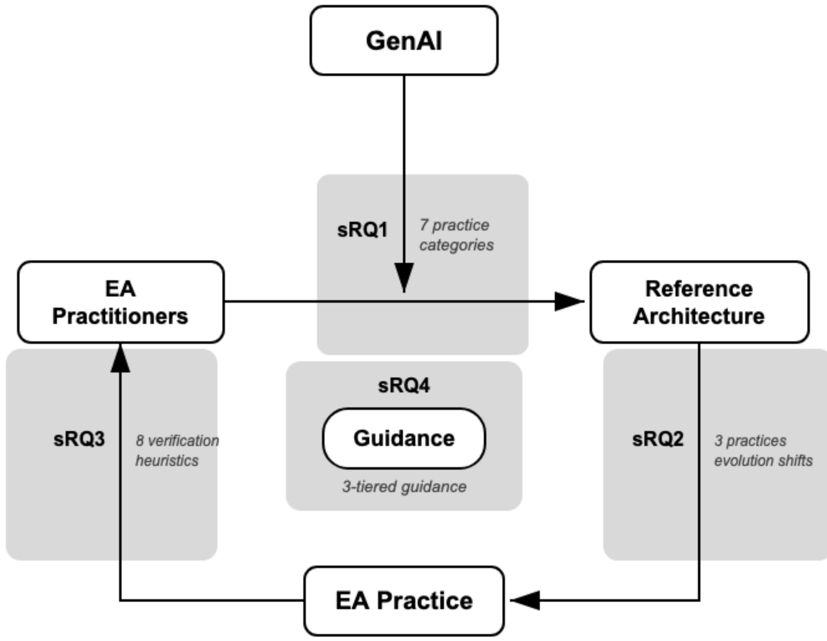

Figure 5. Research conclusions mapped to the analytical framework (cf. Figure 1).

heavily on local conditions.

Third, the guidance artifact offers graduated entry points with embedded traceability mechanisms (verification heuristics, stakeholder-fit checks, and trust calibration references) at each practice level. Practitioners new to GenAI integration can begin with directive-level practices where consensus is strong, then progress to adaptive practices as they develop calibration skills. This structure addresses the guidance vacuum identified in the *Introduction* (Section 1): the absence of evidence-based understanding, evaluation frameworks, risk management guidance, and shared vocabulary for GenAI-augmented EA practice.

Fourth, the gate-to-guardrail governance transition addresses a practical problem: when anyone with GenAI can produce architectural artifacts, periodic review boards cannot keep up. Encoding architectural constraints as machine-readable rules and checking them continuously gives organizations a governance pathway that scales with GenAI-accelerated artifact production.

Fifth, the anti-atrophy strategies address a concrete risk: practitioners lose skills they stop exercising, and GenAI automates precisely the skills (cold-start writing, exhaustive research, cover-to-cover reading) through which architectural expertise develops (Kabashkin, 2025). Three friction strategies (AI-free design sessions, attempt independently first, pair programming mindset) help practitioners maintain the independent capability that makes trust calibration possible.

7.3 Limitations

Limitations (Section 8.7) detailed six categories of limitations qualifying these findings. Three merit brief restatement here so the conclusion stands independently. First, the Delphi panel comprised eight to ten participants drawn from the researcher's professional network, limiting generalizability and introducing self-selection bias toward early adopters. Second, all findings represent a temporal snapshot from January to March 2026; GenAI capabilities evolve rapidly, and specific tool-dependent practices may shift within months. Third, practice data is self-reported through workbooks and facilitated sessions rather than directly observed, introducing potential recall and social desirability biases. One additional qualification bears emphasis: these are early-adopter findings, producing a portrait of frontier practice that may not generalize to the broader EA practitioner population as adoption matures.

8 Discussion

Introduction

This section interprets the findings from *Results and Analysis* (Section 5) and the guidance artifact from *Guidance Artifact* (Section 6) against the theoretical framework established in the *Theoretical Background* (Section 2) and *Conceptual Model* (Section 3). Three lenses structure the interpretation: distributed cognition, practice theory, and the jagged technological frontier. The section then triangulates the practice catalog against external evidence, draws implications for Enterprise Architecture (EA) practice and theory, and closes with limitations and future research directions. The following subsections detail this interpretation:

8.1 Distributed Cognition in EA-GenAI Practice

The conceptual model positioned EA-GenAI integration as a distributed cognitive system where the functional properties of the whole cannot be reduced to the properties of individual agents (Hutchins, 1995). The Delphi findings confirm this framing while revealing specific distribution patterns that the theoretical framework anticipated but did not specify.

8.1.1 Cognitive Distribution Patterns Across Practices

The seven validated practice categories represent distinct configurations of cognitive distribution between human and artificial agents. These configurations align with Hollan et al. (2000)'s three types of distribution (across group members, between internal and external structures, and through time) though the boundaries are more fluid than the typology implies.

The Delphi data suggest that System/Codebase Analysis distributes information retrieval to GenAI while retaining interpretive judgment with the practitioner. The tight consensus on this practice (Table 7) suggests that cognitive distribution succeeds most clearly when the delegated task (scanning code repositories, mapping dependencies) is verifiable against physical systems. The trust spectrum for GenAI in RA development (Table 15) places information compression at the highest trust level, consistent with this interpretation. This convergence implies that verifiability, not task complexity, determines where cognitive distribution succeeds most readily.

Research & Exploration appears to distribute information gathering across a broader knowledge space. The prevalence of Research Assistant mode among Round 1 practices (Section 5.2.1) suggests that practitioners default to using GenAI to compress large bodies of information into actionable summaries. This corresponds to Hollan et al. (2000)'s coordination between internal and external structures: the practitioner's architectural mental model interacts with GenAI's encoded knowledge to produce synthesis neither could achieve alone.

In a similar vein, First Draft & Documentation appears to distribute expression while retaining architectural thinking. The widespread adoption of the Compression Not Generation framing (Section 5.2.3) captures this distinction precisely. As P3 stated: "The thinking is still mine; the AI accelerates expression" (R1 Workbook). GenAI handles the cognitive labour of transforming ideas into structured documents; the practitioner retains the conceptual architecture that gives those documents meaning.

Stakeholder Communication and Governance/Compliance occupy a different position in the distribution landscape. Both exhibited high variability (Table 7), and both involve organizational context that the panel near-unanimously identified as outside GenAI capability (Section 5.2.3). These findings suggest that cognitive distribution encounters a boundary at the organizational context interface; a point we return to below.

8.1.2 The Organizational Context Boundary

The most consistent finding across the Delphi study was the organizational context boundary. The panel near-unanimously identified team dynamics, political constraints, and historical decisions as knowledge that cannot be delegated to GenAI (Section 5.2.3). P3 observed that “organisational context is nearly impossible to convey” (R1 Workbook), while P4 noted that GenAI “is not valuable for the sociotechnical aspects that determine whether an architecture will actually succeed” (R1 Workbook).

In our view, this boundary maps directly to Hutchins (1995)’s concept of functional system boundaries. A distributed cognitive system operates within constraints set by its participants’ capabilities and the media available for coordination. Organizational context (tacit, political, embedded in history) resists being translated into forms GenAI can process. The context engineering practices described in Section 6.3.2 narrow this boundary by making more organizational knowledge machine-accessible, but they do not eliminate it.

This interpretation aligns with Saint-Louis (2019)’s three contexts of EA practice. The technological context maps readily to cognitive distribution (System/Codebase Analysis). The socio-technical context permits partial distribution with human oversight (First Draft & Documentation, Adversarial Review). The eco-technological context (external standards, compliance, regulatory relationships) demands the most careful verification (Governance/Compliance). An alternative explanation is that this boundary reflects current GenAI limitations rather than a structural feature; future systems with richer organizational memory may narrow it. However, the consistency across all three EA contexts suggests the boundary reflects a structural feature of distributed cognition in sociotechnical systems, not solely a technological shortcoming.

8.1.3 Verification as Cognitive Coordination

The universal verification mandate (all eight participants described explicit verification practices, with none accepting GenAI outputs without review (Section 5.2.3)) takes on new significance when viewed through the distributed cognition lens. Verification is not merely quality control. It is the mechanism through which the distributed cognitive system maintains coherence.

Hutchins (1995) described how navigation teams use redundant representations and cross-checking procedures to detect and correct errors propagating through the cognitive system. The Delphi panel's verification practices serve an analogous function. The top-tier heuristics (cross-validation, incremental trust building, and context management; Table 11) are cognitive coordination mechanisms that ensure distributed outputs align with the practitioner's architectural intent.

The verification burden shift, rated the most significant practice change (Table 8), reflects a fundamental restructuring of the cognitive system. Practitioners spend less time on information gathering and expression (now distributed to GenAI) and more time on coordination and validation. This shift is consistent with Hollan et al. (2000)'s observation that distributed cognitive systems require coordination overhead proportional to their distribution complexity.

8.2 Practice Evolution Through a Practice Theory Lens

The conceptual model drew on Feldman and Orlikowski (2011)'s conceptualization of practice as recurrent, recognizable patterns of action that emerge through repeated performance. The Delphi findings provide empirical grounding for how such patterns emerge under conditions of rapid technological change.

8.2.1 From Creator to Curator: A Sociomaterial Accomplishment

The creator-to-curator role transformation (Section 5.3.1) represents what Orlikowski (2007) would describe as a sociomaterial accomplishment. The role shift did not arise from a deliberate redesign of practice. It emerged through situated interaction between practitioners and GenAI capabilities.

Orlikowski (2000)'s concept of technology-in-practice illuminates this process. Practitioners did not adopt GenAI according to a predefined workflow. Instead, they enacted new practices through ongoing interaction, discovering that their most productive contribution shifted from producing architectural content to evaluating, editing, and validating it. Round 2 confirmed this shift: verification burden received the highest significance rating, while the creator-to-curator transformation showed notably higher variability (Table 8).

A selection bias caveat applies: practitioners who gravitate toward curation may be those who adopted GenAI earliest. However, the variability in the creator-to-curator rating is itself informative. Feldman and Orlikowski (2011) emphasize that practices are constituted through performance. The uneven adoption of the curator role across the panel suggests that this transformation is still being constituted, with different organizational contexts producing different enactments.

8.2.2 Emergent Practice Patterns

The seven practice categories exhibit the characteristics Feldman and Orlikowski (2011) identify as constitutive of practice: recurrence, recognizability, and emergence through repeated performance. None of the 19 Round 1 practices were formally designed. All arose through practitioner experimentation with GenAI capabilities in the context of architectural work. This emergence aligns with the conceptual model's prediction that "practices cannot be predicted from GenAI's technical capabilities alone" (Section 3).

The Delphi convergence process itself demonstrated how individual practices aggregate into collective patterns. Round 1 captured 19 individual practices; Round 2 synthesized these into seven validated categories; Round 3 refined heuristics and cross-role dynamics. This progression mirrors the individual-collective dimension identified in the conceptual framework (Section 3.2.1): personal experimentation aggregated into shared organizational patterns through collective deliberation.

Kotusev (2018)'s finding that successful EA practices diverge substantially from mainstream literature prescriptions resonates with this emergence pattern. The seven practice categories did not map neatly onto existing EA frameworks or prescribed GenAI workflows. They emerged from the complex interplay of practitioner creativity, organizational constraints, and technological capabilities; precisely the sociomaterial dynamics that practice theory predicts.

The cross-role dynamics data reinforce this interpretation. All six cross-role patterns in Table 9 exhibited variability between 37% and 79% (Section 5.3.2), indicating that practices crossing traditional role boundaries are still actively being constituted. The "Ghost Architect" provocation surfaced a genuine tension: GenAI democratized artifact production, yet accountability for architectural decisions remained with the architect. This asymmetry (distributed production with concentrated accountability) illustrates the uneven pace at which different dimensions of practice emerge. Feldman and Orlikowski (2011) argue that practices are constituted through performance; when new actors enter the performance space, the practice itself must be reconstituted. The high variability across cross-role patterns suggests that this reconstitution is underway but far from settled, shaped more by organizational context than by GenAI capability alone.

8.2.3 Compression Not Generation as Practice Reframing

The Compression Not Generation pattern represents a collective reframing of GenAI's role in architectural practice, adopted by the majority of the panel. As P2 described: "The value is not in generation but in compression. It compresses reading time for surveys" (R1

Workbook). This framing is significant from a practice theory perspective because it reveals how practitioners make sense of a new technology in terms compatible with their existing professional identity.

Orlikowski and Gash (1994)'s concept of technological frames helps explain this dynamic. By framing GenAI as a compression tool rather than a generative agent, practitioners maintained their self-understanding as the source of architectural knowledge. The technology became an accelerator of human thinking rather than a substitute for it. This framing connects directly to the role identity preservation pattern: all eight participants viewed GenAI as an efficiency tool, not a replacement for professional judgment (Section 5.2.3).

8.3 The Jagged Frontier in Architectural Work

Dell'Acqua et al. (2023) introduced the concept of a "jagged technological frontier" to describe how AI capabilities vary unpredictably across tasks. Their study of Boston Consulting Group consultants found that task performance improved substantially when tasks fell inside the frontier but degraded when they fell outside it. The Delphi consensus–variability profiles map onto this concept in revealing ways.

8.3.1 Mapping the Frontier Through Consensus Data

System/Codebase Analysis sits clearly inside the frontier (Table 7). The tight consensus and low variability suggest that organizational context matters less: codebase analysis is relatively standardized, verifiable, and bounded. This stable frontier boundary suggests that directive guidance may be feasible.

Research & Exploration and Adversarial Review occupy the frontier's middle zone, with moderate consensus supporting contextual guidance, namely general recommendations with situational qualification. The frontier here depends on the specificity of the research question and the practitioner's domain expertise.

Stakeholder Communication and Governance/Compliance sit on or outside the frontier. Their high variability reflects that GenAI's value for these practices depends heavily on organizational context; precisely the knowledge that the panel identified as outside GenAI capability. The shifting frontier boundary suggests that adaptive guidance may be more appropriate for these practices.

Deep Workflow Integration presents a different pattern. Its low mean rating and one abstention likely reflect adoption barriers rather than low frontier placement. The participants operating in Cyborg mode (Section 5.2.1) demonstrated that practitioners who overcome infrastructure prerequisites achieve sustained workflow benefits. The frontier for this practice may be wide but the path to reaching it is steep. This suggests that for Deep Workflow Integration, the primary barrier is organizational readiness rather than GenAI capability.

8.3.2 Frontier Learning Through Failure

The majority of participants described learning GenAI's limitations through errors: hallucinated benchmarks, non-existent folder references, bounded context suggestions that were "technically logical but organisationally nonsensical" (P3, R1 Workbook). These incidents demonstrate what Dell'Acqua et al. (2023) describe as the cost of operating outside the frontier: degraded performance that can propagate downstream if undetected.

In our view, the frontier learning pattern reveals a distinctive feature of the EA-GenAI cognitive system. Unlike the BCG consultants in Dell'Acqua et al. (2023)'s study, who were experimentally assigned to tasks inside or outside the frontier, EA practitioners must navigate the frontier continuously in their daily work. The verification heuristics refined across three Delphi rounds (Table 11) represent collectively developed navigation strategies for this frontier.

The tier structure of verification heuristics reinforces this interpretation. Process-oriented heuristics (cross-validation, incremental trust building, context management) rated higher than experience-dependent ones (domain expertise threshold, learn through failure). This suggests that the panel valued systematic frontier navigation strategies over individual intuition; a finding consistent with Hutchins (1995)'s emphasis on procedural coordination in distributed cognitive systems.

8.3.3 Trust Calibration as Frontier Navigation

Trust calibration (the bidirectional process of adjusting expectations about GenAI capabilities) functions as the primary mechanism for navigating the jagged frontier. The "junior colleague" mental model recurred across the Round 3 panel: "Think of AI as a junior architect and yourself as the senior architect" (R3 Brainstorm). This framing establishes a calibration anchor that adapts as practitioners accumulate experience with GenAI's capability profile.

The panel confirmed that trust calibration is teachable but requires domain expertise as a prerequisite (Section 5.4.3). This finding has implications for the frontier's temporal dimension. Novice practitioners lack the domain knowledge to recognize when GenAI operates outside the frontier, making them more vulnerable to the "beautiful but wrong" outputs that Dell'Acqua et al. (2023) identified. The experience indicators in the guidance artifact

(Section 6.2) address this by providing graduated entry points for practitioners at different experience levels.

8.4 External Triangulation of Delphi Practices

The guidance artifact in Section 6 is grounded in the Delphi study. This section examines how external evidence corroborates, extends, or qualifies the seven validated practice categories. The purpose is triangulation: establishing whether the Delphi findings reflect broader practitioner experience or remain idiosyncratic to this panel.

System/Codebase Analysis. Practitioner evidence confirms codebase analysis as the highest-confidence GenAI application. Practitioners report using GenAI for reverse-engineering software architecture, producing dependency graphs and cross-repository documentation (Chatlatanagulchai et al., 2025). Agent context files containing architecture content appear in 67.7% of analyzed repositories, indicating widespread adoption of this practice in tooling-integrated workflows. This convergence strengthens the directive guidance level assigned to this practice.

Research & Exploration. Multi-model research pipelines are emerging, where practitioners use different AI models for planning, structuring, and execution stages (Banh et al., 2025). The compression framing (GenAI compresses existing information rather than creating new knowledge) is well-established in practitioner discourse and aligns with the Delphi finding that five of eight participants adopted this framing (Section 5.2.3). External evidence thus corroborates both the practice and its underlying cognitive model.

Adversarial Review / Stress Testing. The adversarial use of GenAI is documented across practitioner discourse, though primarily as informal validation rather than structured architectural critique (S. Lee et al., 2025; Stanimirovic et al., 2026). Epistemic stress testing (tracking confidence levels in AI-mediated architectural decisions) connects to the distributed cognition lens by making cognitive assumptions explicit (Gilda & Gilda, 2026). The stress testing techniques identified in Table 14 draw on both Delphi data and this external evidence.

Governance/Compliance. The gap between governance aspiration and implementation is well-documented. Ettinger (2025) identifies significant limitations in existing EA frameworks for addressing GenAI requirements, noting tension between fostering innovation and maintaining compliance. Architecture-as-Code approaches (encoding architectural constraints as machine-readable rules in CI/CD pipelines) represent an emerging response to this gap (Bucaioni et al., 2025). The high variability in the Delphi rating (Table 7) is consistent with this literature: governance value depends heavily on organizational maturity.

First Draft & Documentation. This practice has the strongest external corroboration. Practitioners report generating architecture decision records from meeting transcripts and requirements, with multi-layer verification workflows combining metaprompts, fresh LLM contexts, and LLM-as-Judge evaluation before human review (Banh et al., 2025). The verification burden shift (the highest-rated change theme, Table 8) is most directly experienced in documentation workflows. A related emerging practice, Spec-Driven Development (SDD), extends the documentation-first principle by using formal specifications as the primary human-authored artifact from which implementations derive (Piskala, 2026). SDD is not yet a validated Delphi practice, but its specification-anchored approach mirrors how RAs function: normative specifications constraining downstream implementation. Early evidence suggests three maturity levels (spec-first, spec-anchored, and spec-as-source), though concerns about specification drift and tool inflexibility remain (Piskala, 2026, sec. IX).

Stakeholder Communication. External evidence for GenAI-assisted stakeholder communication in architecture contexts is sparse compared to other practices. The strongest evidence concerns meeting transcript synthesis into formal records (collective decision capture received the highest cross-role pattern rating, Table 9) rather than proactive stakeholder communication. This evidence gap is consistent with the practice's high variability and adaptive guidance level.

Deep Workflow Integration. The shift from chat-based interaction to IDE-integrated workflows represents a maturation trajectory documented across practitioner discourse (Chatlatanagulchai et al., 2025; Rajasekaran et al., 2025). Agent context files containing architecture content appear in 67.7% of studied repositories, and enterprise organizations are converting internal APIs into MCP servers at scale. Context engineering (designing the information environment for AI agents) is emerging as a core architectural skill (Section 6.3.2).

Taken together, external evidence corroborates the Delphi findings across all seven practice categories, though with varying strength. System/Codebase Analysis and First Draft & Documentation have the strongest external validation, consistent with their higher consensus ratings. Stakeholder Communication and Deep Workflow Integration have sparser external evidence, consistent with their adaptive guidance levels. This pattern suggests that the Delphi panel's consensus–variability profiles reflect genuine differences in practice maturity rather than panel idiosyncrasy.

8.5 Implications for EA Practice

The theoretical interpretations above yield four practical implications for how EA practice may evolve as GenAI integration matures.

8.5.1 Context Engineering as an Emerging Competency

The Delphi findings position context engineering (systematically designing the information provided to GenAI systems) as an emerging EA competency. The strong panel endorsement of the context management heuristic (Table 11), combined with the finding that agent context files containing architecture content appear in 67.7% of studied repositories (Chatlatanagulchai et al., 2025), indicates that context engineering is not merely aspirational but reflects emerging practice maturity. These files function as both human-readable documentation and machine-executable constraints.

Context engineering extends the traditional EA practitioner competency set identified by Land et al. (2009) (which emphasized technical expertise, communication skills, and political acumen) with a new dimension. That dimension is the ability to represent architectural knowledge in forms that GenAI systems can process effectively. This is not merely a technical skill. It requires the architectural judgment to determine which constraints matter, the communication skill to encode them precisely, and the organizational awareness to know what context the AI lacks.

8.5.2 The Gate-to-Guardrail Transition

The governance adaptation themes in Section 6.4 suggest a shift from periodic gate reviews to continuous guardrail enforcement. GenAI's acceleration of artifact production (practitioners producing drafts in minutes rather than days) challenges the temporal assumptions of traditional architectural review boards. The Delphi panel's low rating for inside-out governance (Table 9) suggests that most organizations have not yet confronted this shift.

Ettinger (2025) identifies significant limitations in existing EA frameworks for addressing GenAI requirements. In our view, the gate-to-guardrail transition represents the governance equivalent of cognitive distribution: encoding architectural constraints as machine-readable rules distributes governance enforcement across the development pipeline rather than concentrating it in periodic human reviews. This transition directly addresses the accountability gap exposed by the Ghost Architect dynamic (Section 5.3.2): when artifact production is distributed but accountability remains concentrated, continuous guardrail enforcement shifts evaluation from *who produced* the artifact to *whether it meets encoded standards*. In this way, the gate-to-guardrail model resolves the asymmetry between distributed production and concentrated accountability by making compliance verifiable regardless of the artifact's origin.

8.5.3 The Competency Decay Paradox

The skills being automated (cold-start writing, exhaustive manual search, cover-to-cover documentation reading) are the same skills through which practitioners built their expertise (Kabashkin, 2025). The panel rated "skills used less" as the second-most significant change (Table 8), confirming that practitioners recognize this dynamic. Yet "new skills emerging" received a notably lower rating, suggesting practitioners perceive the shift as extending existing expertise rather than requiring fundamentally new competencies.

This tension reveals a paradox that practice theory helps illuminate. If practices are constituted through performance (Feldman & Orlikowski, 2011), then automating the performance undermines the mechanism through which expertise develops. Future practitioners may lack the foundation required for effective trust calibration; precisely the skill the panel identified as essential for all seven practices.

The anti-atrophy strategies proposed in Section 6.3.3 constitute a practice-theoretic response to this paradox. If expertise develops through performance (Feldman & Orlikowski, 2011), then preserving expertise requires deliberately reintroducing the performance that automation displaces. Each strategy targets a different facet of this mechanism. AI-free design sessions preserve foundational skill performance by requiring practitioners to produce architectural work without GenAI assistance. The attempt-independently-first protocol maintains mental model formation by ensuring practitioners engage with problems before encountering AI-generated representations. The pair programming mindset sustains active cognitive engagement during GenAI-assisted work, preventing delegation from becoming abdication. These strategies represent deliberate friction: a structured reintroduction of the cognitive effort that constitutes architectural expertise.

8.5.4 The Maturity Model as Diagnostic

The four-stage maturity model (Table 16) received partial empirical support through the asynchronously collected SP2 maturity progression vote (Table 10). The panel strongly endorsed the natural progression assumption while rejecting the proposition that short experience can substitute for sustained engagement (Table 10). The model's value remains diagnostic rather than prescriptive. Practitioners can use it to locate their current practice and identify potential growth areas without implying that progression through all four stages is necessary or desirable for every organizational context.

The SP2 results reveal a nuanced position that echoes a broader debate in EA maturity literature. Kotusev (2018) demonstrated that EA maturity models often prescribe trajectories that bear little resemblance to how practices actually evolve. The panel's moderate, high-variability agreement on per-category measurement and context-dependence aligns with this critique: progression is real but not uniform across practice categories. The high variability across organizational contexts (particularly in Governance/Compliance and Stakeholder Communication) further suggests that context-specific factors shape adoption pace more than any universal progression logic. The specific four-stage structure remains researcher-proposed, and the asynchronous collection mode introduces methodological caveats (Section 5.6).

8.6 Implications for Theory

The findings extend three theoretical areas: distributed cognition, practice theory, and the conceptualization of GenAI in professional knowledge work.

8.6.1 GenAI as Cognitive Partner

Distributed cognition theory (Hutchins, 1995) originally examined how cognitive processes distribute across human team members and physical artifacts. The Delphi findings suggest that GenAI occupies a position in the cognitive system distinct from both. The organizational context boundary, near-unanimously identified as a hard limit on delegation (Section 5.2.3), delineates where GenAI's cognitive partnership reaches its current limits. Yet within that boundary, practitioners described GenAI generating novel representations, responding to queries contextually, and adapting outputs to conversational framing; capabilities that exceed those of the charts, instruments, and procedures in Hutchins (1995)'s navigation studies. In our view, GenAI functions as a new category of cognitive partner: more capable than a static artifact but less reliable than a human collaborator in domains requiring tacit organizational knowledge.

These findings point toward a pattern not explicitly addressed in Hollan et al. (2000)'s three types of distribution: distribution to agents whose reliability varies across task types. The consensus–variability profiles from Section 8.3 offer preliminary indications of this pattern. The tight consensus on System/Codebase Analysis suggests relatively stable perceived reliability, while the wide disagreement on Stakeholder Communication suggests less predictable performance. These variability differences reflect panel-level variation in importance ratings rather than direct measures of GenAI reliability. Yet the pattern they trace is consistent with the jagged frontier (Dell'Acqua et al., 2023), where AI capability varies unpredictably across task boundaries. While the present study's scope does not permit theoretical generalization, the observation warrants further investigation. Distributed cognition theory, as originally formulated, assumed relatively stable capabilities among system components. GenAI may introduce a partner whose competence shifts across task boundaries, requiring continuous calibration; a dynamic the original framework did not anticipate.

8.6.2 Practice Emergence Under Rapid Technological Change

Feldman and Orlikowski (2011)'s practice theory provides a lens for understanding how recurrent patterns of action emerge and stabilize. The Delphi findings illustrate a distinctive feature of practice emergence under rapid technological change: the technology's capabilities shift faster than practices can stabilize.

The seven practice categories validated in Round 2 showed no challenges or revisions in Round 3 (Section 5.3.3), suggesting that the practice taxonomy itself has stabilized. However, the specific enactments within each category (the tools used, the verification strategies applied, the interaction patterns adopted) continue to evolve as GenAI capabilities advance. This creates what we might describe as stable categories with fluid content: the practice patterns are recognizable but their instantiation changes continuously.

Orlikowski (2000) argued that technology-in-practice emerges through situated use rather than following predetermined adoption paths. The Delphi findings support this claim but add a temporal dimension: when the technology itself evolves rapidly, practitioners must continuously re-situate their practices. The frontier learning pattern (the majority of participants discovering limitations through errors) demonstrates this ongoing re-situation in action.

8.6.3 The Junior Colleague Mental Model

The "junior colleague" framing, recurrent across the Round 3 panel, deserves theoretical attention as a novel calibration heuristic. One participant stated: "Think of AI as a junior architect and yourself as the senior architect" (R3 Brainstorm). Another described treating GenAI interaction like a buddy system where "sharing the elements as if it were human betters the end output" (R3 Brainstorm).

This mental model is not a formal theoretical contribution but a practitioner heuristic that performs important cognitive work. It establishes appropriate expectations (capable but requiring supervision), defines the appropriate relationship (mentorship rather than delegation or replacement), and provides a familiar frame of reference for navigating an unfamiliar technology.

In our view, the junior colleague model is theoretically significant because it reveals how practitioners make sense of a novel cognitive partner by drawing on social frameworks from human collaboration. This process (making the unfamiliar familiar through analogical reasoning) represents a sense-making strategy that existing literature on GenAI adoption has not foregrounded. The framing also reveals its own limitations. Unlike a junior colleague, GenAI does not learn from supervision in ways that transfer across sessions. Its failure modes also differ fundamentally from those of a human novice.

8.7 Limitations

Several limitations qualify the findings and their interpretation.

Sample size and composition. The Delphi panel comprised eight to ten participants across three rounds, sufficient for a Delphi study targeting information richness but limiting generalizability. The panel was predominantly European, self-selected through the researcher's professional network, and composed of practitioners already using GenAI. This composition produces a portrait of early adopter practices that may not represent the broader EA practitioner population.

Scope boundary tension. The workbook was framed as "Software Architecture / Solution Design work" rather than "Enterprise Architecture" to capture practitioners who do RA work under various professional labels (see Section 4). This terminological broadening may have admitted practices that sit at the boundary between RA-specific and general architectural work. The workbook's RA lifecycle mapping (Part C) and R2's RA-contextualized validation mitigate this risk, but some practices (e.g., System/Codebase Analysis) may represent general architectural work that is not exclusive to RA development. This is an inherent trade-off: ecological validity (capturing how practitioners actually describe their work) versus strict construct alignment with the research question's "EA practitioners" scope. Future research could isolate RA-specific practices from general architectural practices through comparative studies across different architectural artifact types.

Temporal snapshot. Data collection occurred between January and March 2026, during a period of rapid GenAI capability evolution. Findings represent a snapshot of practice at a specific moment. GenAI capabilities available during the study period (including large context windows, multi-modal reasoning, and agent-based workflows) may shift substantially within months, potentially rendering specific findings (particularly tool-dependent practices like Deep Workflow Integration) outdated.

Partially validated maturity model. The four-stage maturity model (Table 16) was derived from observed variation in the Delphi data. The asynchronously collected SP2 vote (Table 10) provides empirical support for the underlying progression assumption ($M = 4.71$, variability 22%), partially grounding a maturity model within a consensus-based study. However, a residual design-science tension remains: the specific four-stage structure (from Experimental to Organizational Embedding) was not individually validated by the panel, the vote was collected asynchronously without preceding group discussion, and the sample comprised seven participants. Future research should validate the specific stage boundaries and characteristics.

Delphi process constraints. Three rounds may limit the depth of convergence achievable. Round 1's asynchronous format allowed thoughtful individual responses but prevented the real-time probing possible in interviews. Rounds 2 and 3 compressed discussion into 120-minute sessions, with Round 3 Step 15 not executed and Step 16 collected asynchronously. The absence of Step 15 (Future Perspectives) means the temporal dimension and future-oriented perspectives of the conceptual framework remain empirically underexplored.

Researcher positionality. The researcher is an EA practitioner who uses GenAI daily for architectural work. This dual role means the researcher is simultaneously a research subject and the person designing the inquiry. Insider status touched every phase of the research: the discovery workbook drew on practitioner knowledge of EA workflows, coding decisions reflected familiarity with architectural reasoning patterns, the seven practice categories emerged through researcher synthesis of participant data, and the guidance artifact incorporated the researcher's understanding of what practitioners would find actionable. Each of these steps involved interpretive judgments that insider knowledge could shape, creating a structural bias toward validating the significance of GenAI integration.

Assumption management relied primarily on the Delphi method's structural safeguards rather than a rigorous personal reflexivity protocol. We acknowledge that the researcher's actual reflexive practice was less systematic than planned; reflexive notes were captured at key decision points rather than maintained as a continuous journal. Three structural features partially compensated for this gap. Round 1's asynchronous workbook format ensured participants articulated practices in their own terms before researcher coding began. Round 2's live voting and discussion allowed the panel to validate or reject practice categories collectively, not through researcher judgment alone. Round 3's structured brainstorming generated participant responses that were aggregated before researcher synthesis. In our view, these deliberative safeguards provided meaningful protection against insider bias, but we acknowledge that a more disciplined personal reflexivity practice would have strengthened confirmability.

Two reflexive insights merit disclosure. The researcher has experimented with large language models since 2017 and assumed that most EA practitioners would gradually incorporate GenAI over time. The genuine surprise was the speed of conversion: several participants described themselves as recently skeptical, believing EA work was too context-heavy for AI assistance, yet had rapidly become active daily users within months. This challenged the researcher's gradualist assumption and highlighted the threshold-crossing pattern visible in the findings. Regarding the "Ghost Architect" provocation, this stimulus emerged primarily from the data rather than insider projection. Rounds 1 and 2 surfaced unanticipated cross-role dynamics, including developers producing architectural artifacts bottom-up (Section 5.3.2). The provocation was designed to probe these participant-identified tensions in Round 3. We acknowledge that insider status likely influenced the salience assigned to role boundary shifts; an active practitioner may notice such dynamics more readily. However, the provocation's content was anchored in participant-generated themes rather than researcher preconceptions.

GenAI tool dependency. Findings are shaped by the GenAI tools available in 2025–2026, including large language models from major providers, IDE-integrated coding assistants, and early agent-based workflows. The practices, heuristics, and governance considerations identified may not transfer directly to future GenAI paradigms with different capability profiles.

8.8 Future Research

Five directions merit investigation. First, longitudinal research should track how the seven practice categories evolve as GenAI capabilities advance, since the temporal snapshot limitation means that practices documented here may fragment, merge, or transform as tool capabilities shift.

Second, cross-organizational validation of the practice taxonomy would establish whether the categories identified in this early-adopter panel hold across different organizational contexts, governance maturity levels, and industry sectors. Replication studies with larger panels would test whether the consensus patterns observed here generalize beyond this exploratory sample.

Third, the maturity model's progression assumption received strong panel support ($M = 4.71$), but three areas warrant further validation: the specific four-stage structure (from Experimental to Organizational Embedding) was not individually tested, the practice-category-specific nature of maturity suggested by the panel implies different progression trajectories per practice may exist, and the model's applicability beyond the early-adopter population studied here remains unknown.

Fourth, sector-specific RA contexts (healthcare, financial services, government) introduce regulatory and compliance dimensions that may reshape which practice categories dominate and how verification heuristics are calibrated.

Fifth, the anti-atrophy strategies proposed in Section 8.5 require longitudinal validation. While this study identifies the competency decay paradox and proposes deliberate friction interventions, their long-term effectiveness in preserving architectural expertise remains an open empirical question with significant implications for EA education and professional development.

8.9 Closing Statement

The research began with a practice vacuum: EA practitioners were integrating GenAI into RA development without empirical evidence, evaluation frameworks, shared vocabulary, or validated guidance. Kotusev (2018) called for reconceptualizing EA practice to reflect how architects actually work; GenAI has made that call urgent. Through three Delphi rounds, this study maps what practitioners are doing, how their work is changing, and what guidance helps. The practices documented here will evolve as GenAI capabilities advance. What persists is the underlying question: which parts of architectural work to delegate, and which to keep.

A Appendix - Use of Generative AI in Research

This appendix documents what Generative AI (GenAI) tools were used in this research, how they were deployed, and what functions they served.

Four GenAI systems supported this research:

1. Perplexity: For literature discovery and initial source identification
2. Anthropic Claude: For critical analysis of reading notes, validation of guidance artifact outputs, generation of the machine-readable YAML companion,[10] front cover prompt design, and thesis revision assistance
3. Meta Llama 3 (self-hosted instance): For critical analysis of reading notes
4. Google Gemini: For front cover image generation

These tools served six functions:

1. **Efficiency in discovery**: Accelerated identification of relevant sources across disciplines;
2. **Critical examination**: Provided external perspective on blind spots and gaps in interpretation;
3. **Research alignment**: Helped assess literature relevance to evolving research questions;
4. **Artifact validation**: Checked practitioner checklist cards for internal consistency, completeness against Section 6 content, and formatting;
5. **Format generation**: Produced the machine-readable YAML guidance artifact from researcher-authored content;
6. **Image generation**: Produced the front cover image from a researcher-designed concept.

The following sections document each usage in detail.

A.1 Literature Discovery Process

Perplexity helped identify relevant literature beyond traditional academic search engines (Google Scholar) and supervisor recommendations.

Example discovery prompt: *In the context of Software and Infrastructure engineering, are there any peer-reviewed articles/papers that connect Reference Architecture and tools that support it, including AI tools? If so, give me a list with sources. This is for academic research, so explore and reference only journals, thesis and dissertations - skip the blog posts.*

[10] https://github.com/thiagoavadore/thesis_2024-2026_antwerp_thesis_GenAIPracticeVacuum/blob/main/guidance-artifact.yaml

A.2 Literature Review Process

The review process followed a three-stage pattern:

1. **Reading and Note-Taking**: Papers identified through discovery were added to Zotero and read systematically. Notes captured key points, learnings, questions, and findings.
2. **Critical Analysis**: Notes were submitted to GenAI systems for critical examination using the following prompt:
   I've read the paper (attached) and took notes (also attached) by trying to summarize interesting points, learnings, questions and findings. Investigate potential blind spots, contradictions or any type of hand-waving on my notes. Critique my points and avoid agreeing with me for the sake of making me happy. If the article publication date is before your cut training date, investigate what theories and implications it had that are not highlighted in my notes, always giving me link to sources.
3. **Research Alignment**: Once the Problem Statement and Research Questions crystallized, the analysis prompt evolved to include research alignment:
   As part of my Executive MSc I'm researching into this Problem Statement/Research Question (attached). To better investigate it, I've read the paper (attached) and I took notes (also attached) by trying to summarize interesting points, learnings, questions and findings. Investigate potential blind spots, contradictions or any type of hand-waving on my notes. Critique my points and avoid agreeing with me for the sake of making me happy. If the article publication date is before your cut training date, investigate what theories and implications it had that are not highlighted in my notes, always giving me link to sources. And lastly, investigate if the article supports, contradicts or adds new information to the Problem Statement or the Research Question.

A.3 Guidance Artifact Preparation

Two outputs in the appendices involved GenAI assistance:

1. **Practitioner Checklist Cards (C):** The checklist cards were researcher-authored based on Delphi findings and the guidance artifact in Section 6. GenAI (Claude) was used to validate the cards for internal consistency, completeness against Section 6 content, and formatting. GenAI did not generate the card content.
2. **YAML Guidance Artifact:** The machine-readable YAML companion (available in the research repository[11]) was generated by GenAI (Claude) from the researcher-

[11] https://github.com/thiagoavadore/thesis_2024-2026_antwerp_thesis_GenAIPracticeVacuum/blob/

authored Section 6 and checklist card content. The researcher reviewed, corrected, and approved the output. This constitutes a format transformation of existing researcher-authored material, not new content generation.

A.4 Thesis Revision

During the final revision phase, Claude (Anthropic) assisted with mechanical verification tasks: identifying misformatted cross-references after structural reorganization and verifying APA citation consistency. In all cases, the researcher reviewed the identified issues and applied corrections manually.

A.5 Front Cover Image

The researcher designed the front cover concept to represent the thesis theme visually: cognitive work (represented by an anatomical brain) transforming into an architectural blueprint through a verification lens. Three design constraints guided the concept. First, the image had to be monochromatic to avoid dominating the page. Second, the transformation had to show intentional loss and change in the process, reflecting how GenAI outputs require curation rather than blind acceptance. Third, the style had to match the scholarly tone of an academic thesis.

Anthropic Claude translated these constraints into a structured image generation prompt. Google Gemini then generated the image from the following prompt:

Monochromatic grayscale academic illustration, no text, no letters, no words, no numbers. A detailed anatomical human brain on the left side, drawn in fine ink lines, partially dissolving into geometric particles and dots that drift rightward. These particles gradually reassemble into an abstract architectural blueprint floorplan on the right side, shown as precise technical line drawings of rooms, corridors, and structural elements. Between the brain and the blueprint, a translucent magnifying lens floats, refracting the particles passing through it, symbolizing verification and curation. Subtle watercolor ink splashes in light gray behind both elements. Small geometric shapes (circles, hexagons, connection nodes) scattered in the transition zone. Style: scientific illustration meets architectural drawing, similar to academic thesis cover art. White background, no color, purely grayscale ink and pencil aesthetic. High detail, scholarly tone.

A.6 What GenAI Did Not Do

GenAI tools did not:

main/guidance-artifact.yaml

1. Generate original thesis argumentation or theoretical analysis;
2. Make methodological decisions;
3. Conduct analysis of primary research data;
4. Assist with between-round synthesis, including data coding, taxonomy development, or preparation of subsequent round materials;
5. Develop the conceptual framework;
6. Independently interpret supervisor feedback.

All interpretation, synthesis, framework development, and argumentation is the researcher's own work.

B Appendix - Discovery Workbook Instrument

This appendix reproduces the complete Round 1 discovery workbook distributed to participants in January 2026. The workbook comprises seven parts (A–G) that progress from context setting through practice discovery to reflective sensemaking.

Part A: Context Setting

This opening section establishes the participant's GenAI context and experience level:

- Which GenAI tools do you currently have access to? (e.g., ChatGPT, Claude, Copilot, Gemini, internal tools)
- How long have you been experimenting with GenAI in your EA work?
- What triggered you to start using GenAI for Reference Architecture development?
- Describe your organizational context for EA practice

Part B: Practice Archaeology

The core discovery section employs multiple prompts to excavate practices from different angles:

B.1 Genesis Moments

"Think back to the last 3 Reference Architectures you've worked on. Walk through a typical RA development effort. At what points did you think 'maybe GenAI could help here'? What did you actually try?"

B.2 Practice Inventory

"Now systematically review your GenAI use. For EACH distinct way you use GenAI, please document:"

B.2.1 What do you call this practice? (give it a name)
B.2.2 What problem does it solve?
B.2.3 Describe exactly what you do (step-by-step process)
B.2.4 What specific prompts or techniques have you developed?
B.2.5 How did you discover or develop this approach?
B.2.6 What percentage of the time does it work effectively?
B.2.7 What are the typical failure modes?

Participants documented two to five core practices using this structure.

Part C: Lifecycle Positioning

Participants receive a standard RA lifecycle model *without any GenAI annotations*:

1. Stakeholder Analysis & Requirements Gathering
2. Current State Analysis
3. Pattern Identification & Abstraction
4. Target State Design

5. Component Specification
6. Documentation & Communication
7. Validation & Review
8. Evolution & Maintenance

For each phase, participants answer:

- Which of your named practices apply in this phase?
- What percentage of this phase involves GenAI assistance?
- What tasks MUST remain human-performed? Why?
- What surprised you about GenAI's capabilities or limitations here?

Part D: Critical Incidents

"Share 3-5 specific stories where GenAI significantly impacted your RA work (positive or negative)."

For each incident, participants provide:

- **Context**: What were you trying to accomplish?
- **Action**: Exactly what did you do with GenAI?
- **Result**: What happened? (intended and unintended outcomes)
- **Insight**: What did this teach you about using GenAI for RA work?
- **Evidence**: Can you share specific prompts and outputs? (anonymized as needed)

Part E: Impact Reflection

E.1 Time and Effort Analysis

E.1.1 Estimate hours spent on your last RA with vs. without GenAI
E.1.2 Count iteration cycles needed with GenAI assistance
E.1.3 Identify where GenAI saved time
E.1.4 Identify where GenAI required additional time investment

E.2 Quality and Completeness Assessment

"Comparing GenAI-assisted RAs to traditionally developed ones:"

E.2.1 What aspects show improved quality? Why?
E.2.2 What aspects show degraded quality? Why?
E.2.3 What is simply different rather than better or worse?

E.3 Role Evolution

"How has YOUR role as an EA practitioner changed?"

E.3.1 Skills you use more frequently now
E.3.2 Skills you use less frequently now
E.3.3 New skills you've had to develop
E.3.4 What would happen to your practice if GenAI disappeared tomorrow?

Part F: Challenges and Workarounds

“Document obstacles you’ve encountered using GenAI for Reference Architecture development.”

For each challenge:

- Describe the specific problem
- Explain your workaround (if any)
- Identify what capability you wish GenAI possessed
- Note any red flags you’ve learned to recognize

Part G: Emergent Wisdom

“Based on your experience so far:”

- What essential advice would you give an EA colleague starting with GenAI tomorrow?
- What critical mistakes should they avoid?
- What’s your most counterintuitive discovery about GenAI in RA work?
- Where do you predict GenAI use in RA development will be in 12 months?

C Appendix - Practitioner Checklist Cards

This appendix condenses the guidance artifact (Section 6) into twelve scannable checklist cards. Each card traces directly to a specific subsection of the artifact; no new content is introduced. Practitioners may photocopy or print individual cards for use during Reference Architecture (RA) development sessions. For Cards 1–7, practice descriptions and verification patterns carry Delphi consensus; experience indicators (Novice/Advanced) are researcher proposals.

Card 1: System/Codebase Analysis *[Directive]*

Consensus: $M = 4.00$, $SD = 0.82$	**Provenance:** Delphi consensus

When to Use

- Phase 2 (Current State Analysis) of the RA lifecycle
- Legacy system comprehension, dependency mapping, cross-repository analysis

Key Steps

1. Feed existing system artifacts (code, configs, diagrams) into GenAI for analysis
2. Delegate information gathering to GenAI; retain architectural judgment
3. Cross-validate GenAI-generated descriptions against source systems directly
4. Verify that referenced components, services, and dependencies exist in the actual system

Verification Checklist

- ☐ Cross-validate system descriptions against source systems (Trust level: Highest)
- ☐ Confirm referenced folders, components, and services exist physically
- ☐ Verify dependency graphs against actual system topology
- ☐ Spot-check representative samples of information compression outputs

Novice	**Advanced**
Start with well-documented codebases where verification is straightforward	Extend to legacy systems and cross-repository analysis where GenAI's compression value is highest

External corroboration: Section 8.4

Card 2: Research & Exploration *[Contextual]*

Consensus: $M = 3.60$, $SD = 1.17$	**Provenance:** Delphi consensus

When to Use

- Phase 4 (Target State Design) of the RA lifecycle
- Exploring design alternatives, technology options, and architectural trade-offs

Key Steps

1. Use GenAI as a research assistant for information gathering
2. Scope research questions clearly before prompting
3. Treat GenAI research outputs as hypotheses requiring confirmation
4. Verify claims, statistics, and technology recommendations against primary sources

Verification Checklist

- ☐ Cross-validate specific claims against primary sources (heuristic $M = 4.57$; Trust level: Medium-Low)
- ☐ Verify 60–70% of specific claims against primary sources (P2, R1 Workbook)
- ☐ Confirm technology recommendations against official documentation
- ☐ Check statistics and benchmarks for currency and accuracy

Novice	**Advanced**
Focus on well-scoped research questions where outputs can be verified against established sources	Leverage multi-model workflows and iterative refinement cycles

External corroboration: Section 8.4

Card 3: Adversarial Review / Stress Testing *[Contextual]*

Consensus: $M = 3.60$, $SD = 0.97$	**Provenance:** Delphi consensus

Note: Contextual despite low SD (< 1.0) because practice value depends on practitioner domain expertise.

When to Use

- When challenging, critiquing, or stress-testing architectural decisions
- Requires domain expertise as a prerequisite

Key Steps

1. Select a stress-testing technique (see list below)
2. Actively elicit critique; do not accept GenAI's default collaborative stance
3. Evaluate GenAI-generated critiques against your own domain knowledge
4. Recognize that GenAI cannot model strategic adversarial behaviour across organizational boundaries

Stress-Testing Techniques

- AI Sparring Partner: iterative adversarial dialogue challenging design rationale
- Devil's Advocate Prompting: instruct GenAI to argue against the proposed architecture
- Decision Stress Testing: identify blind spots, steelman alternatives, conduct pre-mortem
- AI Pre-Mortem: "The architecture already failed; why?"
- Skeptical Reviewer: persona critique simulating a senior architect review
- Architectural Gap Analysis: completeness analysis against standards
- Epistemic Stress Testing: challenge confidence levels, not decisions themselves

Verification Checklist

- ☐ Confirm domain expertise threshold before relying on adversarial outputs ($M = 3.86$)
- ☐ Distinguish genuine design weaknesses from pattern-matching artifacts
- ☐ Validate contradiction seeding results against known architectural constraints ($M = 3.33$)

Novice	**Advanced**
Start with structured techniques (AI Pre-Mortem, Skeptical Reviewer) that provide clear prompting templates	Develop intuition for when GenAI critique reflects genuine weaknesses versus pattern-matching artifacts

External corroboration: Section 8.4

Card 4: Governance/Compliance Checking *[Adaptive]*

Consensus: $M = 3.50$, $SD = 1.27$	**Provenance:** Delphi consensus

When to Use

- Verifying architectural artifacts against governance standards and compliance requirements
- Checking organizational policies; adapt to your specific governance structures

Key Steps

1. Scan documents against regulatory requirements using GenAI
2. Verify completeness and accuracy manually in all compliance contexts
3. Apply line-by-line verification for compliance-sensitive outputs ($M = 4.20$)
4. Recognize that compliance errors carry disproportionate organizational risk

Verification Checklist

- ☐ Verify every compliance-related detail line-by-line (Trust level: Very Low)
- ☐ Confirm GenAI output against the actual regulatory text or standard
- ☐ Check for completeness gaps that GenAI may have omitted
- ☐ Validate that organizational-specific policies are correctly represented
- ☐ Verify output structure and formatting against expected standards ($M = 2.83$)

Novice	**Advanced**
Use GenAI for compliance checking only in low-risk contexts initially	Integrate governance checks into automated pipelines where organizational maturity permits

External corroboration: Section 8.4

Card 5: First Draft & Documentation ***[Contextual]***

Consensus: $M = 3.10$, $SD = 1.10$	**Provenance:** Delphi consensus

When to Use

- Producing initial drafts of architectural documentation (solution designs, ADRs, technical specifications, assessment reports)
- Apply the Compression Not Generation framing: GenAI compresses existing information into structured documents

Key Steps

1. Provide existing information (meeting notes, system data, design inputs) as context
2. Use GenAI as a First Draft Generator; treat outputs as starting points, not finished artifacts
3. Cross-check all references, links, and factual claims in generated documentation
4. Apply architectural judgment to refine structure and content

Verification Checklist

- ☐ Cross-check all references and links for accuracy (Trust level: High)
- ☐ Verify factual claims and benchmarks against primary sources
- ☐ Check for hallucinated references or non-existent sources
- ☐ Confirm document structure aligns with organizational templates and standards

Novice	**Advanced**
Start with well-structured document types (meeting minutes, status reports) before progressing to ADRs	Develop metaprompt libraries and multi-stage generation workflows

External corroboration: Section 8.4

Card 6: Stakeholder Communication *[Adaptive]*

Consensus: $M = 3.10$, $SD = 1.37$	**Provenance:** Delphi consensus

When to Use

- Translating architectural content for different audiences (executive summaries, team-specific views, non-technical explanations)
- Synthesizing stakeholder inputs into structured architectural artifacts

Key Steps

1. Define the target audience and communication purpose before prompting
2. Inject organizational context (team dynamics, political constraints, historical decisions) manually
3. Validate that GenAI outputs reflect organizational reality, not just technical logic
4. Review for audience appropriateness before distribution

Verification Checklist

- ☐ Validate organizational context manually before communicating outputs (Trust level: Lowest for organizational judgment)
- ☐ Check that outputs are not "technically logical but organisationally nonsensical" (P3, R1 Workbook)
- ☐ Verify that audience-specific framing preserves architectural intent
- ☐ Confirm that political and interpersonal nuances are appropriately handled
- ☐ Simulate likely stakeholder reactions before presenting GenAI-assisted content ($M = 3.00$)

Novice	**Advanced**
Limit GenAI use to summarization tasks (meeting minutes, document distillation) where source material constrains outputs	Develop audience-specific prompting strategies and RAG-based approaches over organizational documentation

External corroboration: Section 8.4

Card 7: Deep Workflow Integration *[Adaptive]*

Consensus: $M = 2.89$, $SD = 1.17$ (one abstention)	**Provenance:** Delphi consensus

When to Use

- Embedding GenAI into daily workflow through IDE integration, agent context files, custom hooks, MCP servers, and multi-agent orchestration
- Low rating reflects adoption barriers (cognitive load, organizational resistance), not low practice value

Key Steps

1. Begin with single-tool integration (IDE assistant) as a foundation
2. Build trust incrementally; expand GenAI autonomy from low-stakes to higher-stakes tasks ($M = 4.29$)
3. Progress to custom context files and agent orchestration as calibration develops
4. Reduce human oversight touchpoints only after sustained positive calibration

Verification Checklist

- ☐ Apply incremental trust building before expanding automation scope (Trust level: varies by task)
- ☐ Verify automated workflow outputs at regular intervals
- ☐ Confirm that reduced oversight touchpoints are justified by calibration evidence
- ☐ Maintain rollback capability for agentic workflow outputs

Novice	Advanced
Begin with single-tool integration (IDE assistant) before progressing to custom context files and agent orchestration	Design multi-agent architectures with specialized agents for different architectural tasks

External corroboration: Section 8.4

Card 8: Trust Calibration ***Cross-Cutting***

Cross-validation heuristic: $M = 4.57$, $SD = 0.53$	**Provenance:** Researcher synthesis (panel validated heuristics)

Trust Spectrum (from Table 15)

Trust Level	**Task Type**	**Verification Approach**
Highest	Information compression: summaries, boilerplate, dependency graphs	Spot-check representative samples
High	Documentation drafts, concept explanation, test generation	Review structure and key claims
Medium	Code refactoring, common patterns, meeting synthesis	Verify against organizational standards
Medium-Low	Design trade-off exploration, technology recommendations	Cross-validate with primary sources
Low	Architecture decisions, novel business logic	Independent domain expert review
Very Low	Security-critical code, compliance assessment	Line-by-line verification
Lowest	Organizational judgment: politics, team dynamics, legal	Retain entirely; do not delegate

Calibration Checklist

- ☐ Match each GenAI task to its trust level before accepting output
- ☐ Apply the "junior colleague" mental model: supervisory attention for all outputs
- ☐ Recalibrate trust upward when evidence warrants (bidirectional calibration)
- ☐ Build domain expertise first; trust calibration is teachable but requires domain knowledge as a foundation
- ☐ Learn GenAI capability boundaries through direct experience with errors (frontier learning pattern)

Source: Section 6.3.1

Card 9: Context Engineering for EA ***Cross-Cutting***

Context management heuristic: $M = 4.14$, $SD = 0.90$	**Provenance:** Researcher synthesis (panel validated context management)

Agent Context Files

- Encode architectural standards, design principles, and organizational constraints in project-level files (e.g., CLAUDE.md, AGENTS.md)
- These files function as both human-readable documentation and machine-executable governance

- ☐ Create and maintain agent context files for each architectural project
- ☐ Review and update context files regularly (empirical benchmark: every one to three days)
- ☐ Encode architectural standards, design principles, and organizational constraints in each context file

Strategic Information Ordering

- Mitigate the "lost in the middle" effect: language models attend less to information in the middle of long contexts

- ☐ Place critical architectural constraints at the beginning and end of context
- ☐ Use modular rules files (separate documents for different architectural concerns)

Context Scope Management

- Include: architectural principles, relevant standards, system-specific technical context, decision history (ADRs)
- Exclude: organizational politics, exhaustive historical documentation, ambiguous or contradictory requirements

- ☐ Verify that context includes relevant constraints and excludes noise
- ☐ Confirm that organizational politics and interpersonal dynamics are handled by the practitioner, not delegated to GenAI

Source: Section 6.3.2

Card 10: Anti-Atrophy Strategies ***Cross-Cutting***

"Skills used less" significance: $M = 3.90$	**Provenance:** Researcher proposal (panel validated atrophy concern)

The Competency Decay Paradox

The skills being automated (cold-start writing, exhaustive manual search, cover-to-cover documentation reading) are precisely the skills through which practitioners built their expertise. Three deliberate friction practices mitigate this risk.

Friction Practices

- ☐ **AI-Free Design Sessions:** Periodically conduct architectural work without GenAI assistance to maintain foundational skills. *Frequency: schedule regular sessions (analogous to manual flight training despite autopilot capabilities).*
- ☐ **Attempt Independently First:** Engage with a problem for 15–30 minutes before consulting GenAI. This ensures the practitioner develops their own mental model before seeing an AI-generated one. *Frequency: every new design problem.*
- ☐ **Pair Programming Mindset:** Treat GenAI interaction as collaborative work rather than delegation. The practitioner drives architectural thinking; GenAI handles information retrieval and expression. *Frequency: every GenAI session.*

Self-Assessment

- ☐ Can you still perform core architectural tasks without GenAI assistance?
- ☐ Do you form your own mental model before consulting GenAI?
- ☐ Are you actively driving architectural thinking during GenAI sessions?
- ☐ Do you track GenAI failures to refine your trust calibration? (learn through failure, $M = 3.40$)

Source: Section 6.3.3

Card 11: Governance Adaptation ***Organizational***

Evidence base: Thinner Delphi evidence than practice catalog	**Provenance:** Mixed (see per-theme indicators)

Gate-to-Guardrail Transition *(Researcher synthesis)*

- ☐ Assess whether periodic architectural review boards (ARBs) keep pace with GenAI-accelerated artifact production
- ☐ Consider transitioning from periodic gate reviews to continuous guardrail enforcement
- ☐ Encode architectural constraints for automatic checking rather than episodic evaluation
- ☐ Recognize that inside-out governance rated low ($M = 2.00$); most organizations have not yet confronted this shift formally

Architecture-as-Code Enforcement *(Researcher proposal)*

- ☐ Encode architectural constraints as machine-readable rules (TypeScript validators, fitness functions, CI/CD checks)
- ☐ Enable GenAI agents to read and apply architectural rules before generating code
- ☐ Implement executable ADRs where architectural decisions become automated rules in build pipelines
- ☐ Bridge Governance/Compliance practice with Deep Workflow Integration

AI Disclosure and Provenance *(Researcher synthesis)*

- ☐ Establish AI disclosure policies that flag AI-generated components for calibrated review
- ☐ Implement provenance tracking for compliance-sensitive domains
- ☐ Address the "Ghost Architect" dynamic: identify which components were AI-generated or AI-augmented

Source: Section 6.4

Card 12: Maturity Model *Diagnostic*

Progression assumption: $M = 4.71$ (Delphi consensus)	**Provenance:** Researcher proposal (progression assumption has Delphi consensus)

Stage Diagnostic (from Table 16)

Stage	Label	Characteristics	Observable Indicators
1	Experimental	Ad-hoc GenAI use for individual tasks; chat-based interaction; no structured verification	Occasional use for research or drafting; no shared practices; trust calibration through trial and error
2	Systematic Personal	Structured personal workflows; context engineering basics; deliberate verification	Personal prompt libraries; agent context files; anti-atrophy practices; consistent trust spectrum application
3	Team-Level Integration	Shared practices across team; team-level agent context files; Architecture-as-Code adoption	Shared context files in version control; team verification standards; collective decision capture workflows
4	Organizational Embedding	Governance adaptation; enterprise context engineering; agentic workflows	Automated architectural guardrails; AI disclosure policies; multi-agent orchestration; enterprise MCP infrastructure

Key Principles

- ☐ Use this model as diagnostic (locating current practice), not prescriptive (mandating progression)
- ☐ Recognize that progression requires sustained engagement; short experience cannot substitute for long experience ($M = 2.86$)
- ☐ Accept that advancement may occur at different rates across practice categories (per-category measurement $M = 3.86$)
- ☐ Acknowledge that organizational context modulates pace but does not replace experience ($M = 3.86$)

Source: Section 6.5

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